# The Diaz modulus conjecture: a machine-checked spine
and the shape of its obstruction

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**Abstract**

Diaz asked whether a non-zero complex number can be a logarithm of an algebraic number and have algebraic modulus. We give a short formal development of the question. All but two of the numbered statements below have been checked by machine, together with Baker's theorem on linear forms in logarithms, which four of them use, and the two exceptions are flagged where they appear; Appendix A records where each proof lives, so that any claim here can be verified rather than taken.

The results divide in two. The first part is a spine: the strong four exponentials conjecture settles the question outright, by a single instantiation due to Diaz; a statement it implies, which we call $(S)$, settles the part of it that the elementary theory reduces to; and the value that a counterexample to $(S)$ would exhibit cannot be a root of unity. The second part is negative, and is the reason for the paper. We show that one natural line of attack — assembling the certified data of a hypothetical counterexample into a matrix of logarithms with vanishing determinant — provably fails on one branch, that a second branch is circular in a precise sense, and that the remaining branch terminates in a theorem of 1973–74. When this note was first written that theorem had never been formalised; it has been since, in the form the branch needs, and the case it governs is proved with it. The natural interpolation route is blocked as well: no first-order operator with algebraic coefficients has algebraic values there, and the interpolation determinant does not see the candidate's arithmetic at all. The four exponentials conjecture has no instance on the logarithms a counterexample certifies: only a configuration with a constant term can detect one, and for a generic counterexample the period $i\pi$ cannot enter it. What is left is two statements, both open mathematics rather than formalisation, and we say exactly which. A third, which the development had carried until now, turns out not to be needed, and it and the statement it pairs with cannot both be false. Along the way the transcendence theorems now available formally give machine-checked proofs of statements about classical constants, among them that at least one of $\mathrm{e}^{\pi^2}$ and $\mathrm{e}^{i\pi^3}$ is transcendental, a result of Brownawell and Waldschmidt from 1973–74. The smallest open instance of the conjecture, $\sqrt{(\log 2)^2+\pi^2}$, is paired with $2^{i\log 2/\pi}$: at least one of the two is transcendental. Several results below are classical, or special cases of results of Diaz, Roy and Waldschmidt; each is attributed where it appears.

## 1 Setting

Write $\overline{\mathbb{Q}}$ for the algebraic numbers,

$$\mathcal{L} \;=\; \{\lambda \in \mathbb{C} : \mathrm{e}^{\lambda} \in \overline{\mathbb{Q}}\}$$

for the logarithms of algebraic numbers, and $\widetilde{\mathcal{L}}$ for the $\overline{\mathbb{Q}}$-vector subspace of $\mathbb{C}$ spanned by 1 together with $\mathcal{L}$. Complex conjugation is written $^{-}$, and $\overline{\mathrm{e}^{u}} = \mathrm{e}^{\bar{u}}$, so $\mathcal{L}$ is stable under it.

**Conjecture 1.1** (Diaz [17, 1])**.** *For every $u \in \mathbb{C}$ with $u \neq 0$ and $|u| \in \overline{\mathbb{Q}}$, the number $\mathrm{e}^{u}$ is transcendental.*

Waldschmidt's book records the question as open and derives it from the strong four exponentials conjecture below, crediting that observation to Diaz [17, p. 399]; Diaz states it as his Conjecture $C(|u|)$ in [1, §5.1]. His earlier paper [3] treats the homogeneous analogue, in which $|u|^2$ is a rational multiple of $\pi^2$.

We call $u$ a *candidate* when it violates this: $u \neq 0$, $|u| \in \overline{\mathbb{Q}}$, and $\mathrm{e}^u \in \overline{\mathbb{Q}}$. A candidate carries exactly four pieces of certified data,

$$u \in \mathcal{L}, \qquad \bar{u} \in \mathcal{L}, \qquad 2\pi i \in \mathcal{L}, \qquad u\bar{u} = |u|^2 \in \overline{\mathbb{Q}},$$

and the last of these is a *quadratic* relation between logarithms. That is the whole source of the difficulty: the linear theory of logarithms — Baker's theorem and its relatives — says nothing about products, and the theory that does speak about products of two logarithms is the four exponentials circle. Section 2 makes this precise, and everything after it is an attempt to ask for less.

Three classical inputs are used without proof. *Hermite–Lindemann*: for algebraic $a \neq 0$, $\mathrm{e}^a$ is transcendental. *Transcendence of* $\pi$, and hence of $\pi^2$. *Baker's theorem*, in its inhomogeneous form: if $\lambda_1, \dots, \lambda_n \in \mathcal{L}$ are linearly independent over $\mathbb{Q}$, then $1, \lambda_1, \dots, \lambda_n$ are linearly independent over $\overline{\mathbb{Q}}$. Since version 1.16 it is machine-checked too, along the route of Bertrand and Masser [11] through the criterion of Schneider–Lang [17, Ch. 4]. Proposition 5.22, Theorems 6.1 and 6.20 and Corollary 6.21 were first proved with it, or its instance at $u$ and $\bar{u}$, as an explicit hypothesis, and now also without; the case of Waldschmidt's question in Section 2 still carries it, beside Roy's theorem. Three further classical theorems enter in the results added in version 1.1, and all three are machine-checked: the Gel'fond–Schneider theorem, in the formalisation of Karatarakis and Wiedijk [4]; the six exponentials theorem; and the four exponentials theorem in transcendence degree one (Section 5).

**Conjecture 1.2** (Strong four exponentials)**.** *Let $x_1, x_2$ be linearly independent over $\overline{\mathbb{Q}}$ and let $y_1, y_2$ be linearly independent over $\overline{\mathbb{Q}}$. Then at least one of the four products $x_i y_j$ lies outside $\widetilde{\mathcal{L}}$.*

This strengthening of the four exponentials conjecture, with linear independence over $\overline{\mathbb{Q}}$ rather than over $\mathbb{Q}$ and the products allowed to lie in $\widetilde{\mathcal{L}}$, was proposed by Roy [8]; it is Conjecture 11.17 of [17]. Both are open.

## 2 The ceiling

**Theorem 2.1.** *Conjecture 1.2 implies Conjecture 1.1.*

*Proof.* Let $u \neq 0$ with $|u| \in \overline{\mathbb{Q}}$, and suppose $\mathrm{e}^u \in \overline{\mathbb{Q}}$.

If $u$ is algebraic, Hermite–Lindemann gives that $\mathrm{e}^u$ is transcendental, a contradiction. So $u$ is transcendental, and so is $\bar{u}$, since the conjugate of an algebraic number is algebraic. Consequently $(1, u)$ and $(1, \bar{u})$ are each linearly independent over $\overline{\mathbb{Q}}$: a relation $a + bu = 0$ with $a, b \in \overline{\mathbb{Q}}$ not both zero would force either $b \neq 0$ and $u = -a/b \in \overline{\mathbb{Q}}$, or $b = 0$ and $a = 0$.

Apply Conjecture 1.2 to $x = (1, u)$ and $y = (1, \bar{u})$. The four products are

$$1, \qquad \bar{u}, \qquad u, \qquad u\bar{u} = |u|^2,$$

and all four lie in $\widetilde{\mathcal{L}}$: the first by definition of $\widetilde{\mathcal{L}}$; the second and third because $\mathrm{e}^{\bar{u}} = \overline{\mathrm{e}^u}$ and $\mathrm{e}^u$ are algebraic, so $u, \bar{u} \in \mathcal{L}$; and the fourth because $|u|^2$ is algebraic and $\overline{\mathbb{Q}} \cdot 1 \subseteq \widetilde{\mathcal{L}}$. This contradicts Conjecture 1.2. □

The instantiation is the one Waldschmidt's book gives, crediting it to Diaz [17, p. 399]. The same page lists a second open consequence of Conjecture 1.2: if $\lambda \in \mathcal{L} \setminus \{0\}$, $|u| \in \overline{\mathbb{Q}}$ and $\mathrm{e}^{u\lambda} \in \overline{\mathbb{Q}}$, then $u \in \mathbb{Q}$ or $u\lambda/\bar{\lambda} \in \mathbb{Q}$. Its case $u \in \widetilde{\mathcal{L}}$ follows from Roy's strong six exponentials theorem

(Section 6) together with Baker's theorem: the six products of $x = (1, u)$ and $y = (\lambda, \overline{u\lambda}, 1)$ lie in $\widetilde{\mathcal{L}}$, because $u\,\overline{u\lambda} = |u|^2\bar{\lambda}$, so $y$ is linearly dependent over $\overline{\mathbb{Q}}$, and Baker's theorem turns that into $\overline{u\lambda} \in \mathbb{Q}\lambda$. The case $u \in \overline{\mathbb{Q}}$ is Gel'fond–Schneider. This derivation was not found in the sources read; it is machine-checked with both theorems as hypotheses. The full statement follows from Conjecture 1.2, as the book says, and without Baker's theorem: the four products of $x = (1, u)$ and $y = (\lambda, \bar{u}\bar{\lambda})$ lie in $\widetilde{\mathcal{L}}$, and Gel'fond–Schneider, applied to whichever pair is dependent, gives $u \in \mathbb{Q}$ or $\overline{u\lambda} \in \mathbb{Q}\lambda$. That derivation is machine-checked too, with the conjecture as a hypothesis.

**Remark 2.2.** Schanuel's conjecture also suffices, by a different route. If $u$ and $\bar{u}$ are linearly independent over $\mathbb{Q}$, Schanuel at $n = 2$ gives $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}, \mathrm{e}^u, \mathrm{e}^{\bar{u}}) \geq 2$, hence $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}) \geq 2$ once the two exponentials are algebraic; but $u\bar{u} \in \overline{\mathbb{Q}}$ is a non-trivial algebraic relation, so that degree is at most 1. If instead $u = q\bar{u}$ with $q \in \mathbb{Q}$, then $|q| = 1$, so $q = \pm 1$ and $u^2 = \pm|u|^2 \in \overline{\mathbb{Q}}$, making $u$ algebraic; Hermite–Lindemann finishes. Schanuel at $n = 1$ is itself Hermite–Lindemann, so no separate input is needed. This is Diaz's Proposition 1 [1, pp. 550–551], with the same two cases.

**Remark 2.3.** A weaker form already confines the candidates. Kirby's weak Schanuel conjecture [27, Conjecture 1.5] asserts that if $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(a_1, \mathrm{e}^{a_1}, \ldots, a_n, \mathrm{e}^{a_n}) < n$, then $\sum_i m_i a_i \in 2\pi i\mathbb{Z}$ for some integers $m_i$, not all zero. At a candidate $u$ the pair $(u, \bar{u})$ satisfies the hypothesis, as in Remark 2.2, and since a candidate is off both axes, $m_1 u + m_2 \bar{u} \in 2\pi i\mathbb{Z}$ forces $m_1 = -m_2 \neq 0$, hence $\operatorname{Im} u \in \pi\mathbb{Q}$. An integer multiple of $u$ is then a candidate with $\mathrm{e}^u$ real. Assuming it, Conjecture 1.1 is equivalent to statement (ii) of Theorem 3.5, and Conjecture 3.12 is not needed. The weak form says nothing about (ii) itself, nor about Conjecture 3.12: the logarithms involved always satisfy the relation $2 \cdot i\pi \in 2\pi i\mathbb{Z}$. This reduction was not found in the sources read; both of its steps are formal, with only the case $n = 2$ of the conjecture as hypothesis. The name "weak Schanuel" is also used for the algebraic independence of logarithms of algebraic numbers, which is a much stronger statement. Statement (ii) follows in turn from the case $n = 1$ of Schanuel's conjecture over the kernel, $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(\pi, a, \mathrm{e}^a) \geq 2$ for $a \notin 2\pi i\mathbb{Q}$: if $t^2 + \pi^2$ were algebraic, $t$ would be algebraic over $\mathbb{Q}(\pi)$ and $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(\pi, t, \mathrm{e}^t) \leq 1$. So the two hypotheses together give Conjecture 1.1. That step is standard, and the combination is formal; at $a = 1$ the second hypothesis contains the algebraic independence of e and $\pi$, which is open.

Theorem 2.1 fixes the ceiling of the whole subject. Any decomposition of Conjecture 1.1 into cases produces statements that are consequences of Conjecture 1.2, and the useful question is not whether a case follows from it — all of them do — but whether a case follows from something *weaker*. The next section exhibits one such statement.

## 3 Quantisation and the period lattice

This section is self-contained: apart from the transcendence of $\pi$ and of $\pi^2$ it uses no arithmetic input at all. It ends by collapsing an entire branch of the problem to a single sentence about a single real variable.

Throughout, $u$ is off both coordinate axes, meaning $\operatorname{Re} u \neq 0$ and $\operatorname{Im} u \neq 0$.

**Theorem 3.1** (Quantisation)**.** *Let $u$ be off both axes with $\mathrm{e}^u$ real. Then $\operatorname{Im} u \in \pi\mathbb{Z} \setminus \{0\}$, and $|u|^2 > \pi^2$.*

*Proof.* Write $u = t + i\theta$. Then $\mathrm{e}^u = \mathrm{e}^t(\cos\theta + i\sin\theta)$, so $\mathrm{e}^u$ real forces $\mathrm{e}^t \sin\theta = 0$, and $\mathrm{e}^t \neq 0$ gives $\sin\theta = 0$, that is $\theta = k\pi$ for some $k \in \mathbb{Z}$. Since $\operatorname{Im} u \neq 0$ we have $k \neq 0$. Then $|u|^2 = t^2 + k^2\pi^2 \geq t^2 + \pi^2 > \pi^2$, using $t \neq 0$. □

The lower bound is the only effective statement in this development, and it sharpens: if the ratio $\mathrm{e}^u/\overline{\mathrm{e}^u}$ has finite order $m$, the same argument applied to $u/m$ gives $|u|^2 > \pi^2/m^2$. What

that sharpening is worth is settled by the next statement, which says the whole family carries exactly one bit of information.

**Proposition 3.2.** *For $u$ with $\operatorname{Im} u \in \pi\mathbb{Z} \setminus \{0\}$, and with no assumption on $\operatorname{Re} u$, the sharpened bound holds across the entire orbit $\{qu : q \in \mathbb{Q}^\times\}$, that is, $|qu|^2 > \pi^2/m^2$ for every $q \in \mathbb{Q}^\times$ and every $m \geq 1$ with $\left(\mathrm{e}^{qu}/\overline{\mathrm{e}^{qu}}\right)^m = 1$, if and only if $\operatorname{Re} u \neq 0$.*

No algebraicity is used. The point is negative: a predicate that looks like a family of increasingly precise constraints is equivalent to the hypothesis one already has, so no split of the problem along $m$ can be honest.

**Theorem 3.3** (Fibres)**.** *Let $\alpha \in \overline{\mathbb{Q}}^\times$. At most two of the logarithms of $\alpha$ have algebraic modulus.*

*Proof.* The logarithms of $\alpha$ are $u + 2\pi i n$, $n \in \mathbb{Z}$, for any one of them $u$. Writing $\theta = \operatorname{Im} u$,

$$|u + 2\pi i n|^2 \;=\; |u|^2 + 4\pi n\theta + 4\pi^2 n^2.$$

Suppose $n_1, n_2, n_3$ are distinct and all three give an algebraic modulus, hence an algebraic squared modulus. Subtracting the $n_2$ equation from the $n_1$ one and dividing by $4(n_1 - n_2) \neq 0$ gives $\pi\theta + \pi^2(n_1 + n_2) \in \overline{\mathbb{Q}}$, and likewise $\pi\theta + \pi^2(n_1 + n_3) \in \overline{\mathbb{Q}}$. Subtracting these, $\pi^2(n_2 - n_3) \in \overline{\mathbb{Q}}$ with $n_2 \neq n_3$, so $\pi^2 \in \overline{\mathbb{Q}}$ — false. □

**Corollary 3.4.** *If two distinct logarithms of $\alpha$ have algebraic modulus, say $u$ and $u + 2\pi i n$ with $n \neq 0$, then $\pi(\operatorname{Im} u + n\pi) \in \overline{\mathbb{Q}}$.*

*Proof.* Immediate from the displayed identity: the difference of the two squared moduli is $4\pi n\theta + 4\pi^2 n^2$, algebraic, and dividing by $4n$ gives $\pi\theta + \pi^2 n = \pi(\theta + n\pi) \in \overline{\mathbb{Q}}$. □

Corollary 3.4 has no formal counterpart of its own, and Appendix A marks it so. It is worth reading twice. The condition it produces was introduced elsewhere in this subject as a technical case distinction; what it says is that the fibre over $\mathrm{e}^u$ contains a *second* point of algebraic modulus. Theorem 3.3 bounds how often that can happen, and eliminating $\operatorname{Im} u$ between the two displayed relations returns a quartic relation in $\pi$ — which is the reason the class it cuts out has transcendence degree one over $\mathbb{Q}$. The period lattice contributes one bit and no more.

## One sentence

**Theorem 3.5.** *The following are equivalent.*

(i) *For every $u \neq 0$ off both axes with $|u| \in \overline{\mathbb{Q}}$ and $\mathrm{e}^u$ real, $\mathrm{e}^u$ is transcendental.*

(ii) *For every real $t \neq 0$ with $\mathrm{e}^t \in \overline{\mathbb{Q}}$, the number $t^2 + \pi^2$ is transcendental.*

*Proof.* Assume (ii) and let $u$ be as in (i) with $\mathrm{e}^u \in \overline{\mathbb{Q}}$. By Theorem 3.1, $u = t + ik\pi$ with $t = \operatorname{Re} u \neq 0$ and $k \in \mathbb{Z} \setminus \{0\}$. Put $v = u/k$. Then $\mathrm{e}^v$ is a $k$-th root of $\mathrm{e}^u$, hence algebraic; $\operatorname{Im} v = \pi$; $\operatorname{Re} v = t/k \neq 0$; and $|v|^2 = (t/k)^2 + \pi^2$, which is algebraic because $|u|^2$ is and $k$ is rational. Also $\mathrm{e}^v$ is real up to sign, so $\mathrm{e}^{t/k} = \pm\mathrm{e}^v$ is algebraic. Applying (ii) at $t/k$ makes $(t/k)^2 + \pi^2$ transcendental, contradicting the previous sentence.

Conversely assume (i) and let $t \neq 0$ be real with $\mathrm{e}^t \in \overline{\mathbb{Q}}$ and suppose $t^2 + \pi^2 \in \overline{\mathbb{Q}}$. Set $u = t + i\pi$. Then $\mathrm{e}^u = -\mathrm{e}^t$ is real and algebraic, $|u|^2 = t^2 + \pi^2$ is algebraic, and $u$ is off both axes. This contradicts (i). □

The $k$ in that proof carries no content, which is the substance of the theorem: an entire branch of the problem, quantified over two real parameters and an integer, is one assertion about one real variable. Its smallest instance is a question one can state without any of the preceding apparatus.

**Open question.** Is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?

Nothing in this development answers it. Section 2 shows the strong four exponentials conjecture does; Section 4 explains why the available unconditional theory does not reach it. What that theory does give is a partner that must be transcendental if the answer is yes.

**Theorem 3.6.** *Let $t \neq 0$ be real with $\mathrm{e}^t \in \overline{\mathbb{Q}}$, and suppose $\rho = t^2 + \pi^2$ is algebraic. Then $\mathrm{e}^{it^2/\pi}$, $\mathrm{e}^{\pi^2/t}$ and $\mathrm{e}^{\rho/(i\pi)}$ are transcendental.*

*Proof.* The hypothesis makes $t$ algebraic over $\mathbb{Q}(\pi)$, since $t^2 = \rho - \pi^2$; this is its only use. With transcendence degree one available, Proposition 4.4 applies to $(w, z) = (i\pi,\, t/(i\pi))$, where $\mathrm{e}^w = -1$ and $wz = t$, so $wz^2 = -it^2/\pi \notin \mathcal{L}$; and to $(w, z) = (t,\, i\pi/t)$, where $wz = i\pi$, so $wz^2 = -\pi^2/t \notin \mathcal{L}$. Finally $\rho/(i\pi) = -it^2/\pi - i\pi$, so $\mathrm{e}^{\rho/(i\pi)} = -\mathrm{e}^{-it^2/\pi}$. □

**Corollary 3.7.** *At least one of $\sqrt{(\log 2)^2 + \pi^2}$ and $2^{i\log 2/\pi} = \mathrm{e}^{i(\log 2)^2/\pi}$ is transcendental.*

Both statements are special cases of classical results. The proof uses the hypothesis only to make $t$ algebraic over $\mathbb{Q}(\pi)$, and under that weaker hypothesis the first two conclusions follow from a remark of Waldschmidt [15, p. 202]: for $\mathbb{Q}$-linearly independent $\ell_1, \ell_2 \in \mathcal{L}$, either they are algebraically independent or $\exp(\ell_1^2/\ell_2)$ is transcendental. So $2^{i\log 2/\pi}$ is transcendental as soon as $\log 2$ and $\pi$ are algebraically dependent. The second conclusion is also Proposition 1 of [3].

The third number in Theorem 3.6 couples two open statements. The real half of $(S)$, below, asks that $\mathrm{e}^{\gamma/(i\pi)}$ be transcendental for real algebraic $\gamma \neq 0$; the theorem proves it at every $\gamma = t^2 + \pi^2$ that the present question would make algebraic. The two remaining open statements of Section 7 cannot fail at the same number. At rational data they cannot both fail at all.

**Corollary 3.8.** *Let $t \neq 0$ be real with $\mathrm{e}^t \in \overline{\mathbb{Q}}$ and $t^2 + \pi^2 \in \mathbb{Q}$. Then $\mathrm{e}^{i\gamma/\pi}$ is transcendental for every $\gamma \in \mathbb{Q}^\times$.*

*Proof.* Put $r = t^2 + \pi^2$. By Theorem 3.6, $\mathrm{e}^{r/(i\pi)} = \mathrm{e}^{-ir/\pi}$ is transcendental. If $\mathrm{e}^{i\gamma/\pi}$ were algebraic, write $r/\gamma = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$. Then $(\mathrm{e}^{ir/\pi})^b = (\mathrm{e}^{i\gamma/\pi})^a$ would be algebraic, and with it $\mathrm{e}^{ir/\pi}$ and its inverse. □

The statement was not found in the sources read. Its key step is Waldschmidt's remark [15, p. 202] at $\ell_1 = t$ and $\ell_2 = i\pi$, and it is printed as Brownawell's Corollary 5 [12, p. 23] at $\eta = t/\pi$; the scaling step is elementary.

The same argument gives more: a general form, and an instance about a classical constant.

**Proposition 3.9.** *Let $t \neq 0$ be real with $\mathrm{e}^t \in \overline{\mathbb{Q}}$, and let $a, b, c \in \mathbb{Q}$ with $a \neq 0$ and $at^2 + b\pi^2 = c$. Then $\mathrm{e}^{i\gamma/\pi}$ is transcendental for every $\gamma \in \mathbb{Q}^\times$.*

*Proof.* Suppose $\mathrm{e}^{i\gamma/\pi} \in \overline{\mathbb{Q}}$, so that $\lambda = i\gamma/\pi \in \mathcal{L}$. Then $t^2/(i\pi) = -\frac{c}{a\gamma}\,\lambda + \frac{b}{a}\,i\pi$ lies in $\mathcal{L}$, and the matrix $\begin{pmatrix} t & t^2/(i\pi) \\ i\pi & t \end{pmatrix}$ has determinant 0, entries in $\mathcal{L}$, and entries in a field of transcendence degree one, since $t$ is algebraic over $\mathbb{Q}(\pi)$. By the four exponentials theorem in transcendence degree one, its rows or its columns are $\mathbb{Q}$-dependent. Either way $\alpha t + \beta i\pi = 0$ with $\alpha, \beta \in \mathbb{Q}$ not both zero, which is impossible: $t$ is real and non-zero, and $i\pi$ is purely imaginary. □

**Corollary 3.10.** *The number $\mathrm{Li}_2(1/2) = \sum_{n\geq 1} 1/(n^2 2^n) = \pi^2/12 - (\log 2)^2/2$ is irrational, or $\mathrm{e}^{i\gamma/\pi}$ is transcendental for every $\gamma \in \mathbb{Q}^\times$.*

*Proof.* If $\mathrm{Li}_2(1/2) = s \in \mathbb{Q}$, apply Proposition 3.9 with $t = \log 2$, $a = -1/2$, $b = 1/12$ and $c = s$. □

Both alternatives are open. Waldschmidt lists the irrationality of $\mathrm{Li}_2(1/2)$ as unknown [21, p. 274]. It is known for $\mathrm{Li}_2(1/q)$ when $q \geq 6$ or $q \leq -5$ [24, 25], and the case $q = 2$ is still open [26, Remark 2.8.2]. The number $\mathrm{e}^{i/\pi}$ is the smallest case of the real half of the statement $(S)$ below. Proposition 3.9 is Brownawell's Corollary 5 [12, p. 23] at $\eta = t/\pi$, followed by rational scaling; the dichotomy was not found in the sources read.

## A second point in the fibre, and the statement $(S)$

Corollary 3.4 says what it takes for the fibre over $\mathrm{e}^u$ to contain a second point of algebraic modulus. On the branch where the angle is irrational, $\operatorname{Im} u \notin \pi\mathbb{Q}$, that condition does more than constrain $u$: it manufactures a new logarithm.

**Proposition 3.11.** *Let $u \in \mathcal{L}$ with $\operatorname{Im} u \notin \pi\mathbb{Q}$, and suppose $\pi\big(\operatorname{Im} u + r\pi\big) \in \overline{\mathbb{Q}}$ for some rational $r \neq 0$. Then there is a non-zero algebraic $\gamma$ with $\mathrm{e}^{\gamma/(i\pi)}$ algebraic.*

*Proof.* $\mathcal{L}$ is a $\mathbb{Q}$-vector space — $\mathrm{e}^{(a/b)\lambda}$ is a root of $X^b - (\mathrm{e}^{\lambda})^a$ — and it contains $u$, $\bar{u}$ and $2\pi i$. So it contains $\lambda = \frac{1}{2}(u - \bar{u}) + \frac{r}{2}(2\pi i) = i\big(\operatorname{Im} u + r\pi\big)$. Put $\gamma = i\pi\lambda = -\pi\big(\operatorname{Im} u + r\pi\big)$. It is algebraic by hypothesis, and non-zero because $\operatorname{Im} u = -r\pi$ is excluded; and $\gamma/(i\pi) = \lambda \in \mathcal{L}$. □

So on this branch a candidate would exhibit a non-zero algebraic multiple of $1/(i\pi)$ inside $\mathcal{L}$. That is the statement the branch reduces to.

**Conjecture 3.12** ($(S)$)**.** *For every non-zero algebraic $\gamma$, the number $\mathrm{e}^{\gamma/(i\pi)}$ is transcendental. Equivalently, no non-zero algebraic multiple of $1/(i\pi)$ lies in $\mathcal{L}$.*

**Theorem 3.13.** *Conjecture 1.2 implies Conjecture 3.12.*

*Proof.* Suppose $\gamma \neq 0$ is algebraic and $\lambda = \gamma/(i\pi) \in \mathcal{L}$. Apply Conjecture 1.2 to $x = (1, \lambda)$ and $y = (1, i\pi)$. The four products are $1$, $i\pi$, $\lambda$ and $\gamma$, all in $\widetilde{\mathcal{L}}$. Both pairs are $\overline{\mathbb{Q}}$-linearly independent, since $i\pi$ is transcendental and so is $\lambda$, whose algebraicity would make $i\pi = \gamma/\lambda$ algebraic. This contradicts Conjecture 1.2. □

Up to scaling a row, this is the configuration Waldschmidt uses for his Consequence 1.6 in [18], at $\Lambda = i\pi$: under Conjecture 1.2, $1/\pi \notin \widetilde{\mathcal{L}}$, which is stronger than $(S)$.

Roy's strong six exponentials theorem (Section 6) does not give $(S)$, but it puts a price on a failure. An exception to $(S)$ puts $1/\pi$ in $\widetilde{\mathcal{L}}$, and Diaz's Corollaire 1 (PQ) 3) and 4) at $\lambda = i\pi$ [2, p. 380] then keep $\pi^2$ and $1/\pi^2$ out of $\widetilde{\mathcal{L}}$. So under Roy's theorem an exception to $(S)$ would make $\mathrm{e}^{\pi^2}$ transcendental. This is machine-checked with Roy's theorem as a hypothesis.

Whether $(S)$ is strictly weaker than Conjecture 1.2 is not known; what is known is that it asks for much less, one number against a whole configuration. Its hypothetical counterexamples are also more constrained than the value $\mathrm{e}^{\gamma/(i\pi)}$ suggests.

**Proposition 3.14.** *For every non-zero algebraic $\gamma$, the number $\mathrm{e}^{\gamma/(i\pi)}$ is not a root of unity.*

*Proof.* If $\mathrm{e}^{\gamma/(i\pi)} = \mathrm{e}^{2\pi i k/n}$, then $\gamma/(i\pi) = 2\pi i\,(k/n + m)$ for some $m \in \mathbb{Z}$, so $\gamma = -2\pi^2(k/n + m)$. As $\gamma \neq 0$, the rational $k/n + m$ is non-zero and $\pi^2$ is algebraic — false. □

The statement also splits, and not by a case distinction. The set $S_0 = \{\gamma \in \overline{\mathbb{Q}} : \gamma/(i\pi) \in \mathcal{L}\}$ is a $\mathbb{Q}$-vector space, and it is stable under conjugation because $\overline{i\pi} = -i\pi$ gives $\bar{\gamma}/(i\pi) = -\overline{\gamma/(i\pi)}$. So with $\gamma$ it contains $\operatorname{Re}\gamma$ and $i\operatorname{Im}\gamma$, and $(S)$ is equivalent to its two axis halves: for real $\gamma \neq 0$, the number $\mathrm{e}^{-i\gamma/\pi}$, which lies on the unit circle and is not a root of unity, is transcendental; and for $\gamma = i\beta$ with $\beta$ real, $\mathrm{e}^{\beta/\pi}$ is transcendental — that is, $\pi \neq \beta/\log\alpha$ for real algebraic $\beta \neq 0$ and algebraic $\alpha > 0$, $\alpha \neq 1$. The reduction is machine-checked, and both halves are open. They cannot both fail.

**Theorem 3.15** (Diaz)**.** (a) *If $\gamma_1, \gamma_2 \in S_0$ and $\gamma_2 \neq 0$, then $\gamma_1/\gamma_2 \in \mathbb{Q}$.*

(b) *Every element of $S_0$ is real or purely imaginary.*

(c) *At least one of the two halves of $(S)$ holds.*

*Proof.* (a) The numbers $\lambda_j = \gamma_j/(i\pi)$ lie in $\mathcal{L}$, $\lambda_2 \neq 0$, and their ratio $\gamma_1/\gamma_2$ is algebraic. If it were irrational, Gel'fond–Schneider would make $\mathrm{e}^{\lambda_1} = \mathrm{e}^{(\gamma_1/\gamma_2)\lambda_2}$ transcendental. (b) By the conjugation stability of $S_0$ and (a), $\bar{\gamma} = q\gamma$ with $q \in \mathbb{Q}$; comparing real and imaginary parts, either $\operatorname{Re}\gamma = 0$, or $q = 1$ and $\operatorname{Im}\gamma = 0$. (c) A non-zero real element and a non-zero purely imaginary one would be rationally proportional by (a). □

This is Diaz's [1, Théorèmes 4 and 5] at $v = -1/\pi$. He shows that $\mathrm{e}^{v\alpha}$ is transcendental for real $v \neq 0$ and algebraic $\alpha$ off both axes, and that $\{\alpha \in \overline{\mathbb{Q}}^{\times} : \mathrm{e}^{\alpha} \in \overline{\mathbb{Q}}\}$ is empty or of the form $\mathbb{Q}^{\times}\alpha_0$ with $\alpha_0$ on an axis. The proof of the second applies verbatim to $\mathrm{e}^{v\alpha}$, and $\gamma \in S_0$ exactly when $\mathrm{e}^{v\alpha} \in \overline{\mathbb{Q}}$ for $\alpha = i\gamma$.

So $S_0$ is either $\{0\}$ or a single rational line on one axis, and $(S)$ can fail, if at all, in one direction only. The imaginary half has a concrete reading: its exceptions are the $\beta = \pi \log \alpha$ that happen to be algebraic, and by (a) all such $\log \alpha$ are rationally proportional. For instance, at least one of $\pi \log 2$ and $\pi \log 3$ is transcendental, although neither is known to be; this is the transcendence of $\log 3/\log 2$.

Each single instance of $(S)$ also has a partner. For every algebraic $\gamma \neq 0$, at least one of $\mathrm{e}^{i\gamma/\pi}$ and $\mathrm{e}^{i\pi^3/\gamma}$ is transcendental, and at least one of $\mathrm{e}^{i\gamma/\pi}$ and $\mathrm{e}^{i\gamma^2/\pi^3}$ is (Proposition 4.4 with $(w, z) = (\gamma/(i\pi), -\pi^2/\gamma)$ and $(i\pi, -\gamma/\pi^2)$). At $\gamma = 1$: at least one of $\mathrm{e}^{i/\pi}$ and $\mathrm{e}^{i\pi^3}$ is transcendental. These are instances of Corollaire 4 of [13], and the case $\gamma = 1$ is stated there, on p. 192.

Which direction matters for Conjecture 1.1 is settled by looking at what the branch produces. In Proposition 3.11 the number $\gamma = -\pi(\operatorname{Im} u + r\pi)$ is real. The formal decomposition reaches $(S)$ along one other branch, where $\pi \operatorname{Im} u$ itself is algebraic, and there $\gamma = -\pi \operatorname{Im} u$ is real as well. Both branches have been reduced, by machine-checked reductions, to the real half alone. The imaginary half is therefore not needed anywhere in the development (Section 7).

## 4 The precise open boundary

Attach to a candidate $u$, with $\rho = u\bar{u}$ and $r = \sqrt{\rho}$, the matrix

$$H(u, r) \;=\; \begin{pmatrix} u & r \\ r & \bar{u} \end{pmatrix}, \qquad \det H(u, r) = u\bar{u} - r^2 = 0.$$

The matrix is in Waldschmidt's book [17, p. 15], where the ordinary four exponentials conjecture applied to it gives the transcendence of $\mathrm{e}^{|\lambda|}$ for non-real $\lambda \in \mathcal{L}$. Its rows are dependent over $\mathbb{C}$. The next statement says they are dependent in no weaker sense, and it is the sharpest way to see where this problem sits.

**Theorem 4.1** (No vanishing coefficient)**.** *Let $K \subseteq \mathbb{C}$ be a subfield with $r \in K$, $r \neq 0$, and let $u \notin K$ with $u\bar{u} = r^2$. Then for all non-zero $w, v \in K^2$,*

$$w^{\mathsf{T}} H(u, r)\, v \;\neq\; 0.$$

*Proof.* Since $\bar{u} = r^2/u$, the matrix factors as $H(u, r) = (u, r)^{\mathsf{T}}(1, r/u)$, so $w^{\mathsf{T}}Hv = (w_0u + w_1r)(v_0 + v_1r/u)$. If $w_0u + w_1r = 0$, then $w_0 = 0$, since otherwise $u = -w_1r/w_0 \in K$, and then $w_1 = 0$ because $r \neq 0$; so $w = 0$. Likewise $v_0u + v_1r = 0$ forces $v = 0$. (Earlier versions assumed $u$ transcendental over $K$; the factorisation needs only $u \notin K$, as in the Lean statement.) □

At $K = \overline{\mathbb{Q}}$ the transcendence hypothesis is exactly what a candidate supplies: an algebraic $u$ with $\mathrm{e}^u$ algebraic contradicts Hermite–Lindemann, so a candidate is transcendental.

So $H(u, r)$ is singular over $\mathbb{C}$ while every coefficient extracted from it over $\overline{\mathbb{Q}}$ is non-zero. A candidate hides behind exactly this discrepancy: rank drops over $\mathbb{C}$ and nothing detects it over $\overline{\mathbb{Q}}$.

One might hope this configuration is an artefact of a particular choice of matrix. It is not.

**Theorem 4.2** (Uniqueness of the obstruction)**.** *Let $K \subseteq \mathbb{C}$ be a subfield, $u$ transcendental over $K$ with $u \neq 0$ and $u\bar{u} \in K$, and let $A, B, C$ be $2 \times 2$ matrices over $K$ with $\det(A + uB + \bar{u}C) = 0$. Then there is $c \in K$ with*

$$\det(xA + yB + zC) \;=\; c\,(yz - (u\bar{u})\,x^2) \qquad \textit{for all } x, y, z \in \mathbb{C}.$$

Every singular pencil over the candidate's hull is therefore the same object: its determinant is a $K$-multiple of the single quadratic form $yz - \rho x^2$. Concretely, $H(u, r)$ factors as

$$\begin{pmatrix} 0 & r \\ r & 0 \end{pmatrix} (I + uB_0 + \bar{u}C_0), \qquad B_0^2 = C_0^2 = 0, \qquad \operatorname{tr}(B_0 C_0) = \rho^{-1},$$

so the whole family is a nilpotent pencil pinned by one trace. There is no second configuration to look for, and no gain in enlarging the search.

### Descent to a conic, and what is missing

Theorem 4.2 sends the problem to the affine conic

$$X_\rho \;:\; XY = \rho,$$

on which a candidate is the point $(u, \bar{u}) \in \mathcal{L}^2$. This makes the boundary explicit, and it also makes a tempting route collapse.

**Proposition 4.3.** *Fix $\rho \neq 0$ and let $L \subseteq \mathbb{C}$ be any set. Suppose every point of $X_\rho \cap L^2$ lies in a $\mathbb{Q}$-rational vector subspace of $\mathbb{C}^2$ contained in $X_\rho$. Then $X_\rho \cap L^2 = \varnothing$.*

*Proof.* A vector subspace contains the origin, and $0 \cdot 0 = 0 \neq \rho$, so no vector subspace is contained in $X_\rho$. The hypothesis can only be satisfied vacuously. □

The statement is trivial and the reason to record it is what it exposes when read backwards. Roy's conjecture in the shape just used — points of a variety with coordinates in $\mathcal{L}$ lie on rational subspaces contained in it — is available for *homogeneous* quadrics, where there are non-trivial subspaces to be placed on. The affine conic offers none, so the same hypothesis becomes an emptiness statement rather than a structural one, and the gap between the two is not a technicality but the location of the whole difficulty.

Stated in the form that is actually available: Proposition 12.25 of Waldschmidt's book, for arbitrary $k \subseteq K$ and a $k$-subspace $\mathcal{E}$ of $K$, gives the conclusion for a quadric defined over $k$ provided every non-zero matrix $M$ over $\mathcal{E}$ satisfies

$$\operatorname{rank}(M) \;>\; \tfrac{1}{2}\, r_{\mathrm{str}}(M),$$

the structural rank taken over $k$. Structural rank is Roy's notion. What separates this from Diaz's conjecture depends on the setting, and version 1.5 of this note got it wrong.

(a) **Over** $(\overline{\mathbb{Q}}, \mathbb{C}, \widetilde{\mathcal{L}})$, the coefficient field this problem requires, the rank inequality alone would settle it, and homogeneity is no obstacle. The constant 1 lies in $\widetilde{\mathcal{L}}$, so the affine conic is the homogeneous quadric $X_1 X_2 = \rho X_0^2$ at the point $(1, u, \bar{u})$, whose coordinates are linearly independent over $\overline{\mathbb{Q}}$. More directly, a candidate violates the inequality on $H(u, r)$ itself, which has rank 1 and structural rank 2: its structural matrix is $\left(\begin{smallmatrix} X_1 & rX_0 \\ rX_0 & X_2 \end{smallmatrix}\right)$. Roy proved the non-strict bound (Corollary 12.22 of the book). The strict one is open, and for $2 \times 2$ matrices it contains Conjecture 1.2. Equality of rank and structural rank is the Structural Rank Conjecture, equivalent by a theorem of Roy to a case of Schanuel's conjecture [28, 29].

(b) **Over** $(\mathbb{Q}, \mathbb{C}, \mathcal{L})$ the rank inequality is supplied at transcendence degree one by Theorem 15.30 of the same book — which is why the transcendence-degree-one statements in this area are unconditional, and equally why they are not new. There the obstacle is homogeneity: Proposition 12.25 covers homogeneous quadrics, $X_\rho$ is affine, and $\mathcal{L}$ has no constant to homogenise with. Roy and Waldschmidt make the point themselves [7, p. 791]: removing the homogeneity hypothesis from their Théorème 0.2 would give the transcendence of $\mathrm{e}^{\ell^2}$ for non-zero $\ell \in \mathcal{L}$, and that of $\mathrm{e}^{\pi^2}$ is not known (see also [17, p. 593]).

The inhomogeneous direction has been studied. Waldschmidt proved that for a $2 \times 3$ matrix over $\widetilde{\mathcal{L}}$ satisfying suitable independence conditions one of the three $2 \times 2$ minors lies outside $\widetilde{\mathcal{L}}$, and conjectured the same for $2 \times 2$ matrices [19, Theorem 1.2 and Conjecture 1.6]. The matrix $H(u, r)$ meets the hypotheses of that conjecture and has determinant 0, so the conjecture implies Diaz's. Roy extended the Waldschmidt–Masser rank bound to $\mathbb{Q} + \mathcal{L}$ and to $\widetilde{\mathcal{L}}$, and Dasgupta and Kakde expect their Matrix Coefficient Conjecture to extend to $\mathbb{Q} + \mathcal{L}$ [29, §2]. For $2 \times 2$ matrices that extension is the sharp four exponentials conjecture [20, p. 4] with rational constants; it would settle the candidates with $|u|^2 \in \mathbb{Q}$, and Conjecture 3.12 for $\gamma \in \mathbb{Q}$. None of this is known to reach a candidate.

On the unconditional route it is (b) that Theorem 3.5 runs into: $\sqrt{(\log 2)^2 + \pi^2}$ asks about a *sum*, and the transcendence-degree-one theory is homogeneous. The homogeneous companion of that question is settled at transcendence degree one: for non-real $\lambda \in \mathcal{L}$ with $\lambda$, $\bar{\lambda}$ algebraically dependent, $\mathrm{e}^{|\lambda|}$ is transcendental (Diaz [3, Proposition 2]; Roy–Waldschmidt [7, Cor. 7.4], who give this example). So if $\log 2$ and $\pi$ were algebraically dependent, $\exp\sqrt{(\log 2)^2 + \pi^2}$ would be transcendental. Whether $\sqrt{(\log 2)^2 + \pi^2}$ is itself algebraic is the inhomogeneous question, and it is open.

What the homogeneous theory does give at $\mathrm{e}^{\pi^2}$ is worth recording, because it is unconditional and machine-checked. The four exponentials theorem in transcendence degree one (Section 5) applies to the simplest singular matrix whose entries are tied by one ratio.

**Proposition 4.4.** *Let $w \neq 0$ and $z \notin \mathbb{Q}$ with $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(w, z) \leq 1$. Then $w$, $wz$ and $wz^2$ are not all in $\mathcal{L}$.*

*Proof.* Otherwise the matrix $\begin{pmatrix} w & wz \\ wz & wz^2 \end{pmatrix}$ has vanishing determinant and entries in $\mathcal{L}$ generating transcendence degree at most one, so its rows or columns satisfy a non-trivial rational relation $p\,w + q\,wz = 0$; this forces $z = -p/q \in \mathbb{Q}$. □

This is Corollaire 7.4.3 of [15]; an equivalent form is Exercise 15.16(b)(ii) of [17], and it is in [3, p. 241].

**Corollary 4.5** (Brownawell, Waldschmidt)**.** *At least one of $\mathrm{e}^{\pi^2}$ and $\mathrm{e}^{i\pi^3}$ is transcendental.*

*Proof.* Take $w = i\pi$ and $z = -i\pi$: then $\mathrm{e}^w = -1$, $wz = \pi^2$ and $wz^2 = -i\pi^3$. □

This is Corollaire 2 of [13] at $\alpha = -1$, $r = 1$, found independently by Brownawell [12]; Diaz gives this proof [2, p. 386]. Under Roy's strong six exponentials theorem (Section 6) his Conséquence 3 of Corollaire 2 (P) says more at $\lambda = i\pi$: $\pi^2$ and $\pi^3$ are not both in $\widetilde{\mathcal{L}}$ [2, p. 381], so $\mathrm{e}^{\alpha\pi^2}$ or $\mathrm{e}^{\beta\pi^3}$ is transcendental for all algebraic $\alpha, \beta \neq 0$. That is machine-checked with Roy's theorem as a hypothesis. The homogeneous theory says more at the same point: if $\mathrm{e}^{\pi^2}$ is algebraic, then e and $\pi$ are algebraically independent [13, Corollaire 1], and so are $\pi$ and $\mathrm{e}^{i\pi^3}$ [17, Exercise 15.15(c)]. Those need the 1973 theorem in its algebraic-independence form. That theorem is now machine-checked, as an extension of the formal four exponentials theorem of Section 5 to a single algebraic column, and the first of the two statements is formalised with it; the second is not. The problem the 1973 paper takes its title from, Schneider's eighth, is formalised too: at least one of $\mathrm{e}^{\mathrm{e}}$ and $\mathrm{e}^{\mathrm{e}^2}$ is transcendental [13, p. 192].

The same theorem, at $x = (i\pi,\, ir/\pi)$ and $y = (-i\pi/r,\, 1)$, whose column $y_2 = 1$ carries $e^{i\pi} = -1$ and $e^{ir/\pi}$, gives: if $e^{ir/\pi}$ is algebraic for some $r \in \mathbb{Q}^\times$, then two of $\pi$, e and $e^{\pi^2}$ are algebraically independent. The substitution was not found in the sources read, but it is one line from [17, Exercise 15.15(e)]; it is machine-checked, and since version 1.12 unconditionally. At $r = 1$: either $e^{i/\pi}$, the smallest case of the real half of $(S)$, is transcendental, or two of $\pi$, e, $e^{\pi^2}$ are algebraically independent, which is an open problem [17, §15.3.5].

# 5 Two barriers, and one branch that returns

Section 4 located the difficulty. This section records what is known about methods rather than about the problem: two results saying that whole classes of approach cannot work, and one observation about decomposition itself.

### The candidate is invisible to algebraic coefficients

**Theorem 5.1** (Indistinguishability)**.** *Let $u \neq 0$ with $e^u$ algebraic, and let $r \in \overline{\mathbb{Q}}$ be real and non-zero with $u\bar{u} = r^2$. Then there exist a transcendental $t$ with $t\bar{t} = r^2$ and a ring homomorphism $\Phi : \mathbb{C} \to \mathbb{C}$ fixing $\overline{\mathbb{Q}}$ pointwise, carrying $u$ to $t$, commuting with conjugation on $\overline{\mathbb{Q}}(u)$, such that for every matrix $M$ over $\overline{\mathbb{Q}}(u)$ and all vectors $w, v$ over $\overline{\mathbb{Q}}$,*

$$w^{\mathsf{T}} M v = 0 \iff w^{\mathsf{T}} \Phi(M) v = 0.$$

The point $t$ can be any transcendental point of the same circle, for instance $re^i$. It can also be chosen with $e^t$ transcendental, since the circle is uncountable while $\overline{\mathbb{Q}}$ and $\mathcal{L}$ are countable. Whether $re^i$ itself is a candidate is an instance of the conjecture, so version 1.2's "$re^i$ will do" for a non-candidate was too quick. So no vanishing statement with algebraic coefficients, formulated over the candidate's own hull, separates a hypothetical candidate from a point that is certainly not one. Any argument that would refute a candidate by exhibiting such a vanishing relation refutes nothing.

That the intertwining with conjugation comes for free is worth a line: since $\bar{u} = \rho/u$, conjugation is a rational function of the generator on $\overline{\mathbb{Q}}(u)$, so any $\Phi$ matching generators commutes with it automatically. The symmetry is not arranged; it is forced.

The homomorphism of Theorem 5.1 is free to move $\pi$, so a statement whose coefficients involve $\pi$ might hope to see what algebraic coefficients cannot. For a generic candidate it sees nothing either.

**Theorem 5.2** (Indistinguishability over $\mathbb{Q}(\pi)$)**.** *Let $K$ be the algebraic closure of $\mathbb{Q}(\pi)$ in $\mathbb{C}$, and let $u$ be transcendental over $K$ with $|u|^2 \in \overline{\mathbb{Q}}$. Then there exist a point $t$ of the same circle, transcendental over $K$, with $e^t$ transcendental, and a ring homomorphism $\Phi : \mathbb{C} \to \mathbb{C}$ fixing $K$ pointwise, carrying $u$ to $t$, commuting with conjugation on $K(u)$, such that for every matrix $M$ over $K(u)$ and all vectors $w, v$ over $K$,*

$$w^{\mathsf{T}} M v = 0 \iff w^{\mathsf{T}} \Phi(M) v = 0.$$

Being transcendental over $K$ is the same as being algebraically independent of $\pi$, so the hypothesis says that $u$ is generic, in the sense used for Theorem 5.14 below. It cannot be dropped: $\Phi$ fixes $K$. The proof is that of Theorem 5.1 with $K$ in place of $\overline{\mathbb{Q}}$, which works because $K$ is stable under conjugation. The explicit choice $t = re^i$ is no longer available, since it would need $e^i$ transcendental over $\mathbb{Q}(\pi)$; but $K$ is countable, and countability supplies $t$. So nothing algebraic over $\mathbb{Q}(\pi)$ separates a generic candidate from a point that is not a candidate. Only the exponential does: for a candidate, $\Phi$ fixes the algebraic number $e^u$, while $e^{\Phi(u)} = e^t$ is transcendental.

## The threshold is a dimension

**Theorem 5.3** (Dimension count)**.** *Let $u$ be a candidate. Then $1, u, \bar{u}$ all lie in $\widetilde{\mathcal{L}}$ and, granting Hermite–Lindemann, they span an $\overline{\mathbb{Q}}$-subspace of dimension exactly 3. There exist $\overline{\mathbb{Q}}$-independent pairs $x = (x_1, x_2)$ and $y = (y_1, y_2)$ with all four products $x_i y_j$ in that span; and for no $\overline{\mathbb{Q}}$-independent $x$ of length 2 and $y$ of length 3 do all six products $x_i y_j$ lie in it.*

The last clause is a case of a theorem of Roy on spaces of dimension at most three [17, Lemma 12.16 and Exercise 12.10]: a matrix with entries in such a space and linearly independent rows and columns has rank at least a quarter of the number of its rows and columns together, so a $2 \times 3$ one has rank 2.

The certificate a candidate offers is three-dimensional. A $2 \times 2$ configuration needs four products and fits; a $2 \times 3$ configuration needs six and cannot. That is the whole reason the strong four exponentials conjecture reaches this problem while the six exponentials *theorem* — which is proved, and which would otherwise be the natural instrument — does not. The obstruction is not depth but dimension, and no sharpening of six exponentials that stays inside a three-dimensional certificate space can help. Where there is room the theorem works as expected: with $x = (1, \pi)$ and $y = (\log 2, \log 3, \log 5)$ it shows that at least one of $2^\pi$, $3^\pi$, $5^\pi$ is transcendental; and for $\mathbb{Q}$-independent $\lambda, \mu \in \mathcal{L}$, at least one of $\mathrm{e}^{\lambda^2/\mu}$ and $\mathrm{e}^{\mu^2/\lambda}$ is, so that one of $2^{\log_3 2}$ and $3^{\log_2 3}$ is. Both are known in stronger forms: the five exponentials theorem already gives one of $2^\pi$ and $3^\pi$ [16, (2.2.3)], and the second statement is the $\mathbb{Q}$-linear form of [2, Corollaire 4(3)], whose $\widetilde{\mathcal{L}}$ form, under Roy's strong six exponentials theorem (Section 6), is machine-checked as well. The room can also come from a second candidate: Section 6 shows that two candidates of commensurable moduli give the theorem the fourth dimension a single one lacks.

Room helps only where the added numbers are tied to $u$ by an algebraic relation.

**Theorem 5.4.** *Let $u \neq 0$ with $u\bar{u} \in \overline{\mathbb{Q}}$, and let $w_1, \dots, w_m \in \mathbb{C}$ be such that $u, w_1, \dots, w_m$ are algebraically independent over $\overline{\mathbb{Q}}$. Then there are no $\overline{\mathbb{Q}}$-independent $x = (x_1, x_2)$ and $y = (y_1, y_2, y_3)$ with all six products $x_i y_j$ in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \sum_j \overline{\mathbb{Q}} w_j$.*

*Proof.* Multiply by $u$. Since $\bar{u} = \rho/u$ with $\rho \in \overline{\mathbb{Q}}$, every element of the space times $u$ is the value at $(u, w)$ of a polynomial over $\overline{\mathbb{Q}}$ of total degree at most two that is constant once its first variable $X_0$ is set to 0, and evaluation at $(u, w)$ is injective. So a configuration gives a rank-one $2 \times 3$ matrix of such polynomials, which factors as $(h, g) \otimes (c_0, c_1, c_2)$ with both factors free. Total degrees add, and $X_0 \mapsto 0$ is a ring map. If $h$ and $g$ vanish at $X_0 = 0$, they lie in $\overline{\mathbb{Q}}X_0$ unless every $c_j$ is constant. Otherwise every $c_j$ is constant at $X_0 = 0$ and lies in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}X_0$ unless $h$ and $g$ are constant. Each case puts two free vectors in a line or three in a plane. □

The case $m = 0$ is the last clause of Theorem 5.3, and the case $m = 1$, $w_1 = i\pi$ is Theorem 5.16(c) below. The exponential of $u$ plays no role, so the theorem holds for every point of a circle of algebraic radius, candidate or not. For a candidate and logarithms $w_j$ it says that the six exponentials theorem and Roy's strong six exponentials theorem [8], and the sharp form after an application of Baker's theorem, cannot refute the candidate on generic data, however many logarithms are added. Roy [9, §3.2] and Fischler [22, p. 186] make the corresponding point for rank methods and the quadric $xy = z^2$. Where the data satisfy one more relation, Roy's theorem does see something: see Corollaries 6.5 and 6.6 in Section 6.

On a four-dimensional extension of the certificate space, the one further relation can be classified.

**Theorem 5.5.** *Let $u \notin \overline{\mathbb{Q}}$ with $u\bar{u} \in \overline{\mathbb{Q}}$, put $H_0 = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$, and let $z \notin H_0$. There are $\overline{\mathbb{Q}}$-independent $x = (x_1, x_2)$ and $y = (y_1, y_2, y_3)$ with all six products $x_i y_j$ in $H_0 + \overline{\mathbb{Q}}z$ if and only if $z \in H_0 + \overline{\mathbb{Q}}w$ for $w = u^2$, $w = \bar{u}^2$, or $w = 1/(u - a)$ with $a \in \overline{\mathbb{Q}}$, $a \neq 0$.*

*Proof.* Here $u^{-1} = \bar{u}/(u\bar{u}) \in H_0$. If $z \in H_0 + \overline{\mathbb{Q}}w$ and $z \notin H_0$, then $H_0 + \overline{\mathbb{Q}}z = H_0 + \overline{\mathbb{Q}}w$, and the three spaces carry the configurations $(1, u) \otimes (u^{-1}, 1, u)$, $(1, u^{-1}) \otimes (u, 1, u^{-1})$ and $(u^{-1}, (u-a)^{-1}) \otimes (1, u, u^2 - au)$; they are free because $u$ is transcendental. Conversely, let $V = H_0 + \overline{\mathbb{Q}}z$ carry a configuration. Put $h = x_2/x_1$, which is not in $\overline{\mathbb{Q}}$ and so is transcendental, and $B = x_1(\overline{\mathbb{Q}}y_1 + \overline{\mathbb{Q}}y_2 + \overline{\mathbb{Q}}y_3)$. Then $B + hB \subseteq V$ has dimension at most four, so $W = B \cap hB$ and $h^{-1}W$ are subspaces of dimension at least two of the three-dimensional $B$; they meet in some $c = hb \neq 0$ with $b \in B$. Then $b, hb, h^2b \in B$, and $V = b\,(\overline{\mathbb{Q}} + \overline{\mathbb{Q}}h + \overline{\mathbb{Q}}h^2 + \overline{\mathbb{Q}}h^3)$. Write $1 = bP_0(h)$, $u = bP_1(h)$ and $u^{-1} = bP_2(h)$ with $\deg P_i \leq 3$. Since $h$ is transcendental, $P_1P_2 = P_0^2$, and $P_0, P_1$ are independent. Unique factorisation gives $P_0 = gQ_0Q_1$, $P_1 = gQ_1^2$ and $P_2 = gQ_0^2$ with $Q_0, Q_1$ independent of degree at most one and $g = \alpha Q_0 + \beta Q_1$, so $u = Q_1(h)/Q_0(h)$ is a Möbius function of $h$. If $\beta = 0$, then $u^2 = b\,\alpha Q_1(h)^3 \in V$; if $\alpha = 0$, then $u^{-2} = b\,\beta Q_0(h)^3 \in V$, and so $\bar{u}^2 \in V$; otherwise $1/(u + \alpha/\beta) = b\,\beta Q_0(h)^2 Q_1(h) \in V$. The element $w$ found lies in $H_0 + \overline{\mathbb{Q}}z$ but not in $H_0$, so $z \in H_0 + \overline{\mathbb{Q}}w$. □

Theorems 5.4 and 5.5 together say what a rank-one $2 \times 3$ statement can see near a point of a circle: nothing on generic data, and, on a four-dimensional extension of $H_0$, exactly $u^2$, $\bar{u}^2$ and $1/(u-a)$. With Roy's theorem these are the exclusions of Diaz's Corollaire 5(1) and 5(4) [2], behind Corollaries 6.5 and 6.6. The progression $b, bh, bh^2, bh^3$ is the shape of Fischler's Lemma 6.1 [22], for logarithms, and of Diaz's Théorème 7(2) [2], for $\widetilde{\mathcal{L}}$, which use the converse direction: a progression carries a configuration. That a configuration in a four-dimensional space is such a progression is [23, Lemma 4]. The classification itself was not found in the sources read. It does not involve $\mathrm{e}^u$, so it keeps its content if Diaz's conjecture holds. One caveat: $\widetilde{\mathcal{L}}$ is closed under complex conjugation, so a hypothesis $z \in \widetilde{\mathcal{L}}$ brings $\bar{z} \in \widetilde{\mathcal{L}}$ with it, and the space to classify is in general $H_0 + \overline{\mathbb{Q}}z + \overline{\mathbb{Q}}\bar{z}$, of dimension five, where more is excluded. For instance $u^3 \in \widetilde{\mathcal{L}}$ brings $\bar{u}^3$, and $x = (u, \bar{u})$, $y = (1, \bar{u}/u, u/\bar{u})$ is then a configuration: this is Diaz's Corollaire 5(2). Proposition 5.6 treats two such families.

**Proposition 5.6.** *Let $u \notin \overline{\mathbb{Q}}$ with $\rho = u\bar{u} \in \overline{\mathbb{Q}}$, put $H_0 = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$, let $a \in \overline{\mathbb{Q}}$, $a \neq 0$, and let $z = u/(u^2 - a)$ if $a\bar{a} \neq \rho^2$, or $z = u/(u^2 - a)^2$ if $a\bar{a} = \rho^2$. Put $W = H_0 + \overline{\mathbb{Q}}z + \overline{\mathbb{Q}}\bar{z}$.*

(1) *$W$ carries $\overline{\mathbb{Q}}$-independent $x = (x_1, x_2)$ and $y = (y_1, y_2, y_3)$ with all six products $x_iy_j$ in $W$.*

(2) *For no $w \in W \setminus H_0$ does $H_0 + \overline{\mathbb{Q}}w$ carry such a configuration.*

(3) *If $a\bar{a} = \rho^2$, the space $H_0 + \overline{\mathbb{Q}}u/(u^2 - a)$, which is stable under complex conjugation, carries none.*

*Proof.* In the first case put $a' = \rho^2/\bar{a}$, which is algebraic and differs from $a$; then $\bar{z} = -(\rho/\bar{a})\,u/(u^2 - a')$, and let $b = 1/(u(u^2 - a)(u^2 - a'))$. In the second case $\rho^2/\bar{a} = a$ and $\bar{z} = (\rho/\bar{a}^2)\,u^3/(u^2 - a)^2$, and let $b = 1/(u(u^2 - a)^2)$. With $u^{-1} = \bar{u}/\rho$, partial fractions in $u^2$ put $b$, $bu^2$, $bu^4$ and $bu^6$ in $W$, so $x = (1, u^2)$, $y = (b, bu^2, bu^4)$ is a configuration, free because $u$ is transcendental: this is (1). Since $u$ is transcendental, $u \mapsto -u$ is an automorphism of $\overline{\mathbb{Q}}(u)$ over $\overline{\mathbb{Q}}$. It maps $W$ to itself, and the elements of $W$ it fixes are the constants, because $u$, $u^{-1}$, $z$ and $\bar{z}$ are odd. The fixed parts of $u^2$, of $\bar{u}^2 = \rho^2/u^2$ and of $1/(u-c)$ with $c \in \overline{\mathbb{Q}}$, $c \neq 0$, namely $u^2$, $\rho^2/u^2$ and $c/(u^2 - c^2)$, are not constant, so none of the three lies in $W$. If $H_0 + \overline{\mathbb{Q}}w$ carried a configuration, Theorem 5.5 would give $w \in H_0 + \overline{\mathbb{Q}}t$ for one of them, and then $t \in H_0 + \overline{\mathbb{Q}}w \subseteq W$, since $w \notin H_0$: this is (2). For (3), let $z = u/(u^2 - a)^2$; then $u/(u^2 - a) = (\bar{a}^2/\rho)\bar{z} - az$ lies in $W$ and not in $H_0$, and (2) applies. □

**Remark 5.7.** The configuration is not new in shape. With $U = 1/(u^2 - a)$ it is $(u, \bar{u}) \otimes (1, U, \bar{U})$ rescaled, the configuration in the proofs of [1, Théorème 2] and of [2, Théorèmes 6(1) and 7(1)]. At a candidate, Roy's theorem turns (1) into $u/(u^2 - a) \notin \widetilde{\mathcal{L}}$ for $a\bar{a} \neq \rho^2$ and $u/(u^2 - a)^2 \notin \widetilde{\mathcal{L}}$ for $a\bar{a} = \rho^2$. The first is [1, Théorème 2] at $x = (u, \bar{u})$ and $y = U$, since $\bar{u}U = (\rho z - \bar{u})/a$, and is also in [2, Corollaire 4(4) and Théorème 7(1)]; the second is [2, Théorème 6(3)]. Part (2) was not found in the sources read: the configuration is invisible to the four-dimensional spaces of Theorem 5.5.

It does not need all of $W$, since its six products span the odd part $\overline{\mathbb{Q}}u+\overline{\mathbb{Q}}\bar{u}+\overline{\mathbb{Q}}z+\overline{\mathbb{Q}}\bar{z}$. Part (3) is where the hypotheses of those printed results fail for this family, and $\bar{z}$ is what lets the strong six exponentials theorem see $u/(u^2-a)$. A third family, $z=u/(u^2-a_1)+u/(u^2-a_2)$ with $a_1\bar{a}_1=a_2\bar{a}_2=\rho^2$ and $a_1\neq a_2$, behaves in the same way, and the conjugation condition can be dropped altogether: Proposition 5.8.

**Proposition 5.8.** *Let $u\notin\overline{\mathbb{Q}}$ with $\rho=u\bar{u}\in\overline{\mathbb{Q}}$, let $a_1\neq a_2$ be non-zero algebraic numbers, $z_i=u/(u^2-a_i)$, and $W=H_0+\overline{\mathbb{Q}}z_1+\overline{\mathbb{Q}}z_2$.*

(1) *$W$ carries a $2\times3$ configuration, and for no $w\in W\setminus H_0$ does $H_0+\overline{\mathbb{Q}}w$ carry one.*

(2) *If $u$ is a candidate, Roy's theorem gives that $z_1$ and $z_2$ are not both in $\widetilde{\mathcal{L}}$; and if moreover $a_1\bar{a}_1=a_2\bar{a}_2=\rho^2$, then $z_1+z_2\notin\widetilde{\mathcal{L}}$.*

*Proof.* With $b=1/(u(u^2-a_1)(u^2-a_2))$ and $\Delta=a_1-a_2$, partial fractions give $bu^2=(z_1-z_2)/\Delta$, $bu^4=(a_1z_1-a_2z_2)/\Delta$, $bu^6=u+(a_1^2z_1-a_2^2z_2)/\Delta$, and $b$ is a combination of $u^{-1}=\bar{u}/\rho$, $z_1$ and $z_2$; so $x=(1,u^2)$, $y=(b,bu^2,bu^4)$ is a configuration in $W$. Every element of $W$ is a constant plus an odd function of $u$, so the parity argument of Proposition 5.6(2) excludes $u^2$, $\bar{u}^2$ and $1/(u-c)$, $c\neq0$, and Theorem 5.5 with exchange gives (1). For (2), $z_1,z_2\in\widetilde{\mathcal{L}}$ would put $W$ in $\widetilde{\mathcal{L}}$. On the circle $\bar{z}_i=-(\rho/\bar{a}_i)z_i$, with $\rho/\bar{a}_1\neq\rho/\bar{a}_2$, so $z_1+z_2$ and its conjugate together give both $z_i$. □

The configuration is the progression of [2, Théorème 7(2)] with ratio $u^2$ ([22, Lemma 6.1] is its version for logarithms), so part (2) is one substitution there; it is not claimed. The invisibility in (1) was not found in the sources read; it is routine. For $a_2=\rho^2/\bar{a}_1$ the space $W$ is the first family of Proposition 5.6.

Proposition 5.8 is the case $\tau\colon u\mapsto-u$ of a statement about every involution. A Möbius involution $\tau$ of the $u$-line is determined by its *pencil* $\Lambda_\tau$: the quadratics vanishing on its orbits $\{x,\tau x\}$. Conversely every pencil $\Lambda=\overline{\mathbb{Q}}q_1+\overline{\mathbb{Q}}q_2$ with $q_1,q_2\in\overline{\mathbb{Q}}[X]$ coprime, of degree at most 2 and not both of degree below 2, is $\Lambda_\tau$ for exactly one $\tau$.

**Theorem 5.9.** *Let $u\notin\overline{\mathbb{Q}}$ with $\rho=u\bar{u}\in\overline{\mathbb{Q}}$, let $\Lambda=\overline{\mathbb{Q}}q_1+\overline{\mathbb{Q}}q_2$ be such a pencil, $l_1,l_2\in\Lambda$ non-zero, and*

$$W=H_0+\sum_{i=0}^{3}\overline{\mathbb{Q}}\,\frac{q_1(u)^iq_2(u)^{3-i}}{u\,l_1(u)\,l_2(u)}.$$

*Then $W$ carries a $2\times3$ configuration, and contains none of $u^2$, $\bar{u}^2$ and $1/(u-b)$ with $b\in\overline{\mathbb{Q}}^\times$; so no $H_0+\overline{\mathbb{Q}}w\subseteq W$ with $w\notin H_0$ carries one.*

*Proof.* With $r=q_1(u)/q_2(u)$, which is transcendental, $W$ contains the progression $c,cr,cr^2,cr^3$, $c=q_2(u)^3/(u\,l_1(u)l_2(u))$. For the exclusions, write the involution as $\tau(X)=(AX+B)/(\Gamma X-A)$ with $A,B,\Gamma$ the Plücker coordinates of the pencil and $D=A^2+B\Gamma\neq0$; then $(\Gamma u-A)^2q(\tau u)=D\,q(u)$ for $q\in\Lambda$. A membership, cleared of denominators, is a polynomial identity $N(u)L(u)=(u-\beta)(Q(u)L(u)+S(u))$ with $N(\beta)\neq0$. It also holds at $\tau(u)$; eliminating $S$ shows that $\beta$ is a fixed point of $\tau$ and then that $D=0$. The last sentence follows from Theorem 5.5 by exchange. □

**Remark 5.10.** The converse holds. Let $W\supseteq H_0$ be a five-dimensional space of rational functions of $u$ containing a four-term progression $c,ch,ch^2,ch^3$. That a $2\times3$ configuration whose six products span only four dimensions is such a progression is [23, Lemma 4]. If $h$ generates $\overline{\mathbb{Q}}(u)$, then $W$ is a complete linear system $L(E)$ of degree 4 with $E\geq[0]+[\infty]$, and contains $u^2$, $u^{-2}$ or some $1/(u-a)$. Otherwise $\overline{\mathbb{Q}}(h)$ has index 2: the index is at most 2 because two independent elements of $H_0\cap\sum_k\overline{\mathbb{Q}}ch^k$ have a ratio of degree at most 2 in $u$ lying in $\overline{\mathbb{Q}}(h)$. Then $\overline{\mathbb{Q}}(h)$ is the field of invariants of an involution $\tau$, and $W$ is the space of Theorem 5.9 for $\Lambda=\Lambda_\tau$. For a ratio $h$ rational in $u$ all of this is formal, the index bound through the degree formula

$[\overline{\mathbb{Q}}(X):\overline{\mathbb{Q}}(f)]=\max(\deg p,\deg q)$ of Mathlib's proof of Lüroth's theorem; the progression step and the rationality of $h$ stay in the text. So the five-dimensional spaces that carry a progression and are invisible to Theorem 5.5 are exactly those of Theorem 5.9. Configurations whose products fill the whole space are a separate case: $\overline{\mathbb{Q}}u^{-1}+\overline{\mathbb{Q}}+\overline{\mathbb{Q}}u+\overline{\mathbb{Q}}u^3+\overline{\mathbb{Q}}u^4$ carries $(1,u)\otimes(u^{-1},1,u^3)$ and contains none of $u^2$, $u^{-2}$, $1/(u-b)$; it is not stable under conjugation. Conjugation-stable spaces of that kind exist (Proposition 5.11). None of these statements was found in the sources read; at a candidate their exclusions are, as before, substitutions in [2, Théorème 7(2)].

The configurations whose products fill the space, with one factor of degree one, are $(1,u)\otimes(u^{-1},1,g)$ after rescaling; their spaces are $H_0+\overline{\mathbb{Q}}g+\overline{\mathbb{Q}}ug$. Those that conjugation respects form one explicit family.

**Proposition 5.11.** *Let $u\notin\overline{\mathbb{Q}}$, and let $s\in\overline{\mathbb{Q}}$ be real with $s^2=u\bar u$.*

(i) *Let $\lambda\in\overline{\mathbb{Q}}$ with $\lambda\bar\lambda=1$ and $f\in\mathbb{R}$, and suppose $1,u,\bar u,f,g$ are free, where $g=f/(s+\lambda u)$. Then $W(\lambda,f)=H_0+\overline{\mathbb{Q}}f+\overline{\mathbb{Q}}g$ is stable under conjugation and carries the configuration $(1,u)\otimes(u^{-1},1,g)$.*

(ii) *Conversely, if $1,u,\bar u,g,ug$ are free and $\bar g\in\overline{\mathbb{Q}}+\overline{\mathbb{Q}}u+\overline{\mathbb{Q}}ug$, then $H_0+\overline{\mathbb{Q}}g+\overline{\mathbb{Q}}ug$ is one of the spaces $W(\lambda,f)$.*

(iii) *For $\lambda=1$ and $f=u^2/s^2+s^2/u^2$ the five numbers are free and $W(1,f)$ contains none of $u^2$, $\bar u^2$ and $1/(u-b)$, $b\in\overline{\mathbb{Q}}^\times$; so no $H_0+\overline{\mathbb{Q}}w\subseteq W$ with $w\notin H_0$ carries a configuration. It contains no four-term progression.*

(iv) *If $u$ is a candidate, $s=|u|$ and the five numbers of (i) are free, Roy's theorem gives that $f$ and $f/(|u|+\lambda u)$ are not both in $\widetilde{\mathcal{L}}$.*

*Proof.* (i) Since $\bar u=s^2/u$ and $\lambda\bar\lambda=1$, $\bar g=f/s-g$ and $ug=(f-sg)/\lambda$. (ii) Write $\bar g=a+bu+c\,ug$ and conjugate: $(1-c\bar cs^2)g\in\overline{\mathbb{Q}}+\overline{\mathbb{Q}}\bar u$, so $c\bar cs^2=1$, and the constants satisfy $\bar a=-\bar cs^2b$. With $q=a/(cs^2)$, $g'=g+q\bar u$ satisfies $\bar{g'}=c\,ug'$; then $\lambda=cs$ has modulus 1, $f=g'(s+\lambda u)$ is real, and the two spans agree. (iii) With $v=u/s$, modulo $H_0$ the generator $f/(s+u)$ is $s^{-1}(v^{-2}+2/(v+1))$; compare coefficients. Theorem 5.5 and exchange give the rest. The polar divisor $2[0]+2[\infty]+[-s]$ has degree 5, while the spaces of Theorem 5.9 have degree 6. (iv) At a candidate $W(\lambda,f)\subseteq\widetilde{\mathcal{L}}$ would contain the configuration of (i). □

So the conjecture that every conjugation-stable space with a configuration invisible to Theorem 5.5 contains a progression is false. Statements (i)–(iii) were not found in the sources read; they are routine. Statement (iv) is one substitution in [2, Théorème 7(1)], with $v_0=f/(|u|+\lambda u)$; it is also [1, Théorème 2] at $x=(v_0,\bar v_0)$. The two questions this leaves, which $f$ make $W(\lambda,f)$ visible and whether every conjugation-stable invisible space is of this kind, are answered below.

**Proposition 5.12.** *Let $u\notin\overline{\mathbb{Q}}$, and let $s\in\overline{\mathbb{Q}}$ be real with $s^2=u\bar u$.*

(i) *With $F=u^3/s^3+u^2/s^2$, the space $W_1=H_0+\overline{\mathbb{Q}}F+\overline{\mathbb{Q}}\bar F$ is five-dimensional and stable under conjugation, the six products of $(1,u^2)$ and $(\bar u,u+s,\bar F)$ span it, and it contains none of $u^2$, $\bar u^2$ and $1/(u-b)$, $b\in\overline{\mathbb{Q}}^\times$; so no $H_0+\overline{\mathbb{Q}}w\subseteq W_1$ with $w\notin H_0$ carries a configuration. No configuration in $W_1$ has a ratio $x_2/x_1=(\alpha u+\beta)/(\gamma u+\delta)$ with $\alpha\delta-\beta\gamma\neq0$; in particular $W_1$ is none of the spaces $W(\lambda,f)$.*

(ii) *Let $a,b,\alpha,\beta,\gamma,\delta\in\overline{\mathbb{Q}}^\times$ with $a^2\neq b^2$, $g_1=\alpha/(u-a)+\beta/(u+a)$, $g_2=\gamma/(u-b)+\delta/(u+b)$ and $W=H_0+\overline{\mathbb{Q}}g_1+\overline{\mathbb{Q}}g_2$. Then $W$ is five-dimensional and contains none of $u^2$, $u^{-2}$, $1/(u-c)$. If $\alpha\neq\beta$ or $\gamma\neq\delta$, the products of $(1,u^2)$ and $(u^{-1},g_1,g_2)$ span $W$, and no $H_0+\overline{\mathbb{Q}}w\subseteq W$ with $w\notin H_0$ carries a configuration. If $a\bar a=b\bar b=s^2$ and $\beta/\alpha$, $\delta/\gamma$ are real, $W$ is stable under conjugation.*

(iii) *Let $a,a_1,a_2\in\overline{\mathbb{Q}}^\times$ with $a\bar a\neq s^2$ and $a_1\neq a_2$, and let $g=a_1/(u-a)+a_2/(u+a)$. Then $H_0+\overline{\mathbb{Q}}g+\overline{\mathbb{Q}}\bar g$ is stable under conjugation, the products of a configuration span it, and no $H_0+\overline{\mathbb{Q}}w$ inside it with $w\notin H_0$ carries a configuration.*

(iv) *With $v = u/s$, $G_1 = 1/(v-1)+2/(v+1)$ and $G_2 = 1/(v-1)^2+1/(v-1)-2/(v+1)^2+2/(v+1)$, the space $W_2 = H_0 + \overline{\mathbb{Q}}G_1 + \overline{\mathbb{Q}}G_2$ is five-dimensional and stable under conjugation, the six products of $(1, u^2)$ and $(\bar{u}, G_1, G_2)$ span it, and it contains none of $u^2$, $\bar{u}^2$ and $1/(u-e)$, $e \in \overline{\mathbb{Q}}^\times$; so no $H_0 + \overline{\mathbb{Q}}w \subseteq W_2$ with $w \notin H_0$ carries a configuration. No configuration in $W_2$ has a ratio $(\alpha u+\beta)/(\gamma u+\delta)$ with $\alpha\delta - \beta\gamma \neq 0$.*

*Proof.* (i) With $\bar{u} = s^2/u$, $s^3F = u^3 + su^2$ and $u^3\bar{F} = s^3 + s^2u$, a combination of the five generators times $s^3u^3$ is a polynomial of degree 6 in $u$; comparing coefficients gives freeness and the exclusions. The products are $\bar{u}$, $u+s$, $\bar{F}$, $s^2u$, $s^3F$ and $s\bar{u}+s^2$. If $x_2(\gamma u+\delta) = x_1(\alpha u+\beta)$, then $c = \alpha x_1 - \gamma x_2 \neq 0$ and $t_j = cy_j$ satisfies $t_j, ut_j \in W_1$; but such $t$ lie in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\bar{u}$, which cannot hold three free $t_j$. (ii) Residues at the five poles $0, \pm a, \pm b$ give freeness and the exclusions; $u^2g_1 = (\alpha+\beta)u + (\alpha-\beta)a + a^2g_1$ gives the products, which contain 1 when $(\alpha-\beta, \gamma-\delta) \neq 0$. Conjugation sends $1/(u \mp a)$ to $\mp 1/\bar{a} - (a/\bar{a})/(u \mp a)$ when $a\bar{a} = s^2$. (iii) With $b = s^2/\bar{a}$, $\bar{g}$ is a constant plus $\gamma/(u-b) + \delta/(u+b)$, where $\gamma = -\bar{a}_1 b/\bar{a}$ and $\delta = -\bar{a}_2 b/\bar{a}$, and $a^2 \neq b^2$ since $a\bar{a} + s^2 > 0$; so (ii) applies to $g$ and $\bar{g}$, and conjugation permutes the generators. (iv) $\bar{G}_1 = 1 - G_1$ and $\bar{G}_2 = G_2$; $u^2G_1 = 3su - s^2 + s^2G_1$ and $u^2G_2 = s^2G_2 + 3su - 2s^2 + 2s^2G_1$ give the products, and clearing by $u(u^2-s^2)^2$ gives freeness and the exclusions as in (i). An element of $W_2$ with a pole at $s$ has one at $-s$, the leading polar coefficients being in the ratio $1:-2$ for double poles and $1:2$ for simple ones; multiplication by $u$ changes the sign of that ratio, so $t, ut \in W_2$ forces $t \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\bar{u}$, and the argument of (i) applies. In all four parts invisibility is Theorem 5.5 and exchange. □

**Remark 5.13.** For $\alpha = \beta$, $\gamma = \delta$ the family (ii) is that of Proposition 5.8, whose configuration is a progression; so (ii) is its Case II deformation, and its factor $(1, u^2)$ has degree two, while every $W(\lambda, f)$ has a factor of degree one. Factors of higher degree do not occur: if $1, u, u^{-1}$ lie in $c \cdot \mathrm{span}(1, h, h^2, k, hk)$ and $h \notin \overline{\mathbb{Q}}$, then $u$ is a root of a non-zero quadratic over $\overline{\mathbb{Q}}[h]$, so a ratio $h = p(u)/q(u)$ has $\max(\deg p, \deg q) \leq 2$ (formal, by the degree formula of Mathlib's proof of Lüroth's theorem; that a configuration whose products span $W$ gives this shape is in the research notes). For a real Laurent polynomial $f = \sum_{|j|\leq k} c_j(u/s)^j$, $W(\lambda, f)$ is invisible to Theorem 5.5 when $k \geq 3$, or $k = 2$ and $f(-\bar{\lambda}) \neq 0$, and contains $u^2$ and $\bar{u}^2$ when $k = 2$ and $f(-\bar{\lambda}) = 0$ (formal); so the example of Proposition 5.11(iii) is the generic case and $(u/s - s/u)^2$ a visible one. For general $f$, $W(\lambda, f)$ is visible exactly when it is a complete linear system, and the conjugation-stable invisible spaces with a factor of degree two form three families: two orbits of $u \mapsto -u$ fixed by conjugation, two orbits exchanged by it, and a cyclic family, in which $(1, u^2)$ acts modulo $H_0$ as a Jordan block. These two statements are proved in the research notes and are not formalised. Each family has a formal member in Proposition 5.12: (ii) in its conjugation-stable case for the first, (i) (orbits 0 and $\infty$) and (iii) for the second, (iv) for the third. None of this was found in the sources read; it is routine.

## The four exponentials conjecture sees nothing homogeneous

Theorem 5.3 works over $\overline{\mathbb{Q}}$ and counts dimensions. Over $\mathbb{Q}$, the field of the four exponentials conjecture itself, dimension is the wrong invariant: $x = y = (1, \sqrt{2})$ is a $2 \times 2$ configuration inside a two-dimensional space. The right invariant is the absence of homogeneous relations. Call a candidate *generic* if $u$ and $\pi$ are algebraically independent (over $\overline{\mathbb{Q}}$, equivalently over $\mathbb{Q}$); otherwise $u$ is algebraic over $\mathbb{Q}(\pi)$.

**Theorem 5.14.** *Let $u \neq 0$ with $|u|^2 \in \overline{\mathbb{Q}}$, and suppose $u$ and $\pi$ are algebraically independent over $\overline{\mathbb{Q}}$.*

(a) *No non-zero rational quadratic form vanishes at $(u, \bar{u}, i\pi)$.*

(b) *Every singular $2 \times 2$ matrix with entries in $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has $\mathbb{Q}$-dependent rows or $\mathbb{Q}$-dependent columns.*

(c) *The same holds with $\overline{\mathbb{Q}}$ in place of $\mathbb{Q}$: there are no $\overline{\mathbb{Q}}$-independent pairs $x = (x_1, x_2)$, $y = (y_1, y_2)$ with all four products $x_i y_j$ in $\overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \overline{\mathbb{Q}}i\pi$.*

*Proof.* (a) Substitute $\bar{u} = \rho/u$ and multiply by $u^2$. The result is a polynomial in $u$ and $i\pi$ with the six distinct monomials $u^4, u^3 i\pi, u^2(i\pi)^2, u^2, u\, i\pi, 1$ and coefficients the symmetrised coefficients of the form, three of them multiplied by $\rho$ or $\rho^2$. By independence all vanish. (b) The determinant is a rational quadratic form in $(u, \bar{u}, i\pi)$, hence identically zero by (a). A $2 \times 2$ matrix of linear forms with identically vanishing determinant has dependent rows or columns, by unique factorisation in the polynomial ring. (c) By Theorem 5.16(a) below the determinant form is a multiple of $X_1X_2 - \rho X_0^2$; it has no constant coefficient, so the multiple is zero, and the lemma on linear forms, which holds over any field of characteristic zero, applies over $\overline{\mathbb{Q}}$. □

Nothing here uses $\mathrm{e}^u$: the statement is about any point of the circle that is generic in this sense. For a candidate it says that the four exponentials *conjecture*, fed with the logarithms the candidate certifies, has no instance at all. The non-generic candidates are no better targets, for the opposite reason.

**Proposition 5.15.** *Let $u$ be a candidate algebraic over $\mathbb{Q}(\pi)$. Then every singular $2 \times 2$ matrix with entries in $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has $\mathbb{Q}$-dependent rows or $\mathbb{Q}$-dependent columns.*

*Proof.* A zero entry forces a zero row or column. Otherwise all entries lie in $\mathcal{L}$ and are algebraic over $\mathbb{Q}(\pi)$, and the four exponentials theorem in transcendence degree one applies. □

So for every candidate, the four exponentials conjecture restricted to its certified logarithms is already a theorem: by algebra in the generic case, by the 1973–74 theorem otherwise. The same holds for an exception $\lambda = \gamma/(i\pi)$ to $(S)$: every singular $2 \times 2$ matrix over $\mathbb{Q}\lambda + \mathbb{Q}i\pi$ has dependent rows or columns, because $\lambda/(i\pi) = -\gamma/\pi^2$ is transcendental.

In the generic case the same holds in every size for the Matrix Coefficient Conjecture of Dasgupta and Kakde [29, Conjectures 1.2 and 2.3], which at $n = 2$ is equivalent to the four exponentials conjecture: every singular $n \times n$ matrix $M$ with entries in $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has non-zero $v, w \in \mathbb{Q}^n$ with $\langle w, Mv \rangle = 0$, and the same holds over $\overline{\mathbb{Q}}$. The proof combines Roy's lemma on spaces of singular matrices [29, Theorem 2.2], of which [17, Proposition 12.5] is a stronger form, with the absence of homogeneous relations of every degree at $(u, \bar{u}, i\pi)$, an instance of [17, Proposition 12.13]. All three steps are machine-checked. Waldschmidt's book also records the conjecture that its property $\left(\begin{smallmatrix} A & B \\ C & 0 \end{smallmatrix}\right)$ holds for $(\mathbb{Q}, \mathbb{C}, \mathcal{L})$ [17, §12.4.1]; for square matrices that property implies the Matrix Coefficient Conjecture. The same section records the homogeneous form of the conjecture on algebraic independence of logarithms: if $\lambda_0, \ldots, \lambda_n \in \mathcal{L}$ are linearly independent over $\mathbb{Q}$, then $\lambda_1/\lambda_0, \ldots, \lambda_n/\lambda_0$ are algebraically independent. At a generic candidate's own logarithms $u$, $\bar{u}$ and $i\pi$ it holds: that is the absence of homogeneous relations at $(u, \bar{u}, i\pi)$ just used, and the one relation the candidate does satisfy, $u\bar{u} = \rho$, is not homogeneous. So that conjecture cannot refute a generic candidate through its own logarithms either.

What does detect a counterexample needs a constant term. The configurations of Theorems 2.1 and 3.13 have one, and in the generic case the constant is the only thing they can add to $u$ and $\bar{u}$.

**Theorem 5.16** (The period never enters)**.** *Let $u \neq 0$ with $\rho = |u|^2 \in \overline{\mathbb{Q}}$, and suppose $u$ and $\pi$ are algebraically independent over $\overline{\mathbb{Q}}$.*

(a) *Every quadratic form with algebraic coefficients that vanishes at $(1, u, \bar{u}, i\pi)$ is a multiple of $X_1X_2 - \rho X_0^2$.*

(b) *Let $M$ be a singular $2 \times 2$ matrix with entries in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \overline{\mathbb{Q}}i\pi$, whose rows are $\overline{\mathbb{Q}}$-independent and whose columns are $\overline{\mathbb{Q}}$-independent. Then the coefficient of $i\pi$ in every entry of $M$ is zero.*

(c) *There are no $\overline{\mathbb{Q}}$-independent $x = (x_1, x_2)$ and $y = (y_1, y_2, y_3)$ with all six products $x_i y_j$ in $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \overline{\mathbb{Q}}i\pi$.*

*Proof.* (a) As in Theorem 5.14(a), now with nine monomials; the only two terms of the form that land on the same monomial are $X_0^2$ and $X_1X_2$, both at $u^2$. (b) Write the entries as $\sum_k x_k C_k$ at $x = (1, u, \bar{u}, i\pi)$, with constant matrices $C_k$. By (a), $\det(\sum_k x_k C_k) = c\,(x_1x_2 - \rho x_0^2)$ identically, and $c \neq 0$, since otherwise the lemma on linear forms from the proof of Theorem 5.14(b), which holds over any field of characteristic zero and here is used over $\overline{\mathbb{Q}}$, gives dependent rows or columns. For the polar form of the determinant, $C_3$ is then isotropic and orthogonal to $C_0, C_1, C_2$, which span a non-degenerate three-dimensional subspace of the four-dimensional space of $2 \times 2$ matrices. Its orthogonal complement is a non-degenerate line, and the only isotropic vector on it is zero. (c) By (a), each $2 \times 2$ minor of the matrix $(x_i y_j)$, written in the coordinates $1, u, \bar{u}, i\pi$, is a multiple $c_{jl}$ of the norm form. If $c_{01} = 0$, the lemma on linear forms over $\overline{\mathbb{Q}}$ makes $x$ or $(y_1, y_2)$ dependent. Otherwise the identity $D_0L_{i0} - D_1L_{i1} + D_2L_{i2} = 0$, satisfied by the minors $D_j$ of any $2 \times 3$ matrix, makes the norm form times a linear form vanish identically; so that linear form vanishes, and at $(1, u, \bar{u}, i\pi)$ it gives an $\overline{\mathbb{Q}}$-relation among the $y_j$ with a non-zero coefficient. □

The configuration $\left(\begin{smallmatrix} 1 & \bar{u} \\ u & \rho \end{smallmatrix}\right)$ of Theorem 2.1 is of this kind. The logarithm $i\pi$ of $-1$, which every candidate supplies for free, is of no use to it. Part (c) extends Theorem 5.3 from $1, u, \bar{u}$ to $1, u, \bar{u}, i\pi$, where counting dimensions no longer decides: Roy's strong six exponentials theorem [8], in Diaz's words the known result closest to the strong four exponentials conjecture [2, p. 379], gives nothing against a generic candidate even with the period added. Diaz observed that the theorem has produced no example of a $\lambda \in \widetilde{\mathcal{L}}$ with $|\lambda| \notin \widetilde{\mathcal{L}}$ [2]; on a candidate's own data, this is one reason.

Points of a circle can carry homogeneous relations, and these can be listed. Candidates cannot, as the discussion after the proposition shows.

**Proposition 5.17.** *Let $u = x + iy$ with $x \neq 0$ and $y \notin \mathbb{Q}\pi$.*

(a) *The rational quadratic relations among $u$, $\bar{u}$, $i\pi$ are exactly the identities $Q(y, \pi) = s\,|u|^2$, with $Q$ a rational binary quadratic form and $s \in \mathbb{Q}$.*

(b) *Suppose $|u|^2 \in \overline{\mathbb{Q}}$, and that $u$ satisfies such a relation whose ternary form has no non-trivial rational zero. Then every singular $2 \times 2$ matrix over $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has $\mathbb{Q}$-dependent rows or $\mathbb{Q}$-dependent columns.*

(c) *For every real algebraic $\rho > \pi^2$, some $u$ with $|u|^2 = \rho$, $x \neq 0$ and $y \notin \mathbb{Q}\pi$ satisfies $\mathrm{Re}(u^2) = \pi^2$, a relation of the kind in (b).*

*Proof.* (a) Compare real and imaginary parts. (b) The determinant of a $2 \times 2$ matrix of rational linear forms in three variables has a non-trivial rational zero: take a common zero of the first row. By (a) and the transcendence of $\pi$, every relation $u$ carries is a rational multiple of the given one. The determinant of a matrix as in (b) is such a relation, and it has a rational zero while the given form has none, so it vanishes identically, and the lemma on linear forms finishes. (c) Take $x^2 = (\rho + \pi^2)/2$ and $y^2 = (\rho - \pi^2)/2$; the form $\frac{1}{2}(X_1^2 + X_2^2) + X_3^2$ is positive definite. □

None of these relations occurs at a candidate. A candidate algebraic over $\mathbb{Q}(\pi)$ with $y \notin \mathbb{Q}\pi$ has $u$, $\bar{u}$, $i\pi$ linearly independent over $\mathbb{Q}$, and all three are logarithms of algebraic numbers in a field of transcendence degree one. By Théorème 0.2 of Roy–Waldschmidt [7], the analogue of the 1973–74 theorem for an arbitrary quadric, such numbers satisfy no non-trivial rational quadratic relation. With Theorem 5.14(a) for the generic case, no candidate with $\mathrm{Im}\,u \notin \mathbb{Q}\pi$ satisfies a rational quadratic relation among $u$, $\bar{u}$, $i\pi$, and for the points of (c) $\mathrm{e}^u$ is transcendental. This is the transcendence-degree-one statement of §4(a) at work. What Proposition 5.17 adds is narrower, and does not use $\mathrm{e}^u$: a relation with a non-trivial rational zero can be a determinant, and one without, like $\mathrm{Re}(u^2) = \pi^2$, never is, so no $2 \times 2$ configuration sees it. Roy and Waldschmidt

reach it through larger matrices, via the Clifford algebra of the form. Version 1.4 said that such a relation is excluded by nothing this note knows of; that was wrong.

### One mechanism: separation, a normal form, Laurent hulls

The results of this section so far share one mechanism, which works over any subfield $K \subseteq \mathbb{C}$. A $p \times q$ *configuration over* $K$ in a $K$-space $V \subseteq \mathbb{C}$ is a pair of $K$-linearly independent families $x_1, \dots, x_p$ and $y_1, \dots, y_q$ with every product $x_i y_j$ in $V$.

**Theorem 5.18.** *Let $K \subseteq F$ be subfields of $\mathbb{C}$.*

(a) *If $V_0 \subseteq F$ is a $K$-space and $w_1, \dots, w_m$ are algebraically independent over $F$, every $p \times q$ configuration over $K$ with $p, q \geq 2$ in $V_0 + Kw_1 + \cdots + Kw_m$ lies in $V_0$.*

(b) *Let $u$ be transcendental over $K$ with $\rho = u\bar{u} \in K$, and $H_0 = K + Ku + K\bar{u}$. Every $p \times q$ configuration over $K$ in $H_0$ has $p + q \leq 4$, and the $2 \times 2$ ones are exactly*

$$x_i = \mu\,(P_{i1} + P_{i2}u), \qquad y_j = \mu^{-1}(Q_{1j} + Q_{2j}\bar{u}), \qquad \mu \neq 0,\ P, Q \in \mathrm{GL}_2(K)\,:$$

*the orbit of the certificate $(1, u) \otimes (1, \bar{u})$ under $\mathrm{GL}_2(K) \times \mathrm{GL}_2(K)$.*

(c) *If moreover $u, w_1, \dots, w_m$ are algebraically independent over $K$, every configuration over $K$ with $p, q \geq 2$ in $H_0 + Kw_1 + \cdots + Kw_m$ is $2 \times 2$ and lies in $H_0$, and $x_i y_j - c_{ij} \in Ku + K\bar{u}$ for the invertible matrix $c = P\,\mathrm{diag}(1, \rho)\,Q$.*

(d) *At a candidate, with $u, w_1, \dots, w_m$ algebraically independent over $\overline{\mathbb{Q}}$, every row and every column of such a $2 \times 2$ configuration over $\overline{\mathbb{Q}}$ has an entry outside $\overline{\mathbb{Q}}\mathcal{L}$.*

*Proof.* (a) For one number $w$ and $p = q = 2$, write $x_i y_j = c_{ij} + \ell_{ij} w$ with $c_{ij} \in V_0$ and $\ell_{ij} \in K$. The determinant $\det(C + wL)$ vanishes and $w$ is transcendental over $F$, so $\det L = 0$. If a column $j$ of $L$ were non-zero, $z = \ell_{2j}x_1 - \ell_{1j}x_2 \neq 0$ would have $zy_1, zy_2 \in F$, and comparing the coefficients of $w$ in $(zy_1)(x_i y_2) = (zy_2)(x_i y_1)$ gives $z(\ell_{i2}y_1 - \ell_{i1}y_2) = 0$ for each $i$, so $L = 0$ by the independence of $y$. In general, induct on $m$ over $F(w_1, \dots, w_{m-1})$, using that every entry lies in a $2 \times 2$ block. (b) $H_0$ is a subspace of $K(u)$ of dimension 3, and a $p \times q$ configuration in a finite-dimensional $V_0 \subseteq K(u)$ has $p + q \leq \dim V_0 + 1$. To see this, clear denominators: $g(u)x_i y_j = A_{ij}(u)$ with $A_{ij} \in K[X]$ and $A_{i1}A_{1j} = A_{11}A_{ij}$; then the spaces spanned by the $A_{i1}$ and by the $A_{1j}$ have dimensions $p$ and $q$, and the linear Cauchy–Davenport inequality in $K[X]$, proved by counting orders at 0, bounds the space spanned by their products. For the normal form, $ux_i y_j = A_{ij}(u)$ with $\deg A_{ij} \leq 2$, the vanishing minor factors $A_{ij} = a_i b_j$ in $K[X]$, neither the $a_i$ nor the $b_j$ are both constant, so all have degree at most one, and $1/u = \bar{u}/\rho$ gives the form of $y$. (c) By (a) with $F = K(u)$, then (b), expanding with $u\bar{u} = \rho$. (d) By (c) with $K = \overline{\mathbb{Q}}$: an entry with $c_{ij} \neq 0$ in $\overline{\mathbb{Q}}\mathcal{L}$ would put the algebraic number $c_{ij}$ in $\overline{\mathbb{Q}}\mathcal{L}$, since $u, \bar{u} \in \mathcal{L}$, and Baker's theorem gives $\overline{\mathbb{Q}} \cap \overline{\mathbb{Q}}\mathcal{L} = 0$. □

**Proposition 5.19.** *Let $u$ be transcendental over $K$ and, for $S \subseteq \mathbb{Z}$, let $V_S$ be the $K$-span of the $u^s$, $s \in S$.*

(a) *For finite $S$, $V_S$ carries a $p \times q$ configuration iff $A + B \subseteq S$ for some $|A| = p$, $|B| = q$; for $2 \times 3$, iff some difference occurs three times in $S$.*

(b) *For $4 \leq k < l$, $V_{\{0, \pm 1, \pm k, \pm l\}}$ carries a $2 \times 3$ configuration iff $l \in \{k+1, k+2, 2k-1, 2k, 2k+1, 3k\}$.*

(c) *$V_S$ with $S = \{0, \pm 1\} \cup \{\pm 4^j : j \geq 1\}$ carries no $2 \times 3$ configuration.*

*Proof.* (a) Distinct powers of $u$ are independent, which gives one direction. Conversely, write $P_{ij}(u) = u^N x_i y_j$ with $P_{ij}$ supported on $S + N$; the spans of the $P_{i1}$ and of the $P_{1j}$ have $p$ and $q$ distinct orders at 0, and $fg = P_{11}h$ with $h$ in the span of the $P_{ij}$ turns their sums into elements of

$S$ after a shift. For $|A| = 2$, $A = \{a, a+d\}$, and $A+B \subseteq S$ says that $a+B$ lies in $\{s \in S : s+d \in S\}$. (b) The positive differences of $\{0, \pm 1, \pm k, \pm l\}$ are $1, k-1, k, k+1, l-1, l, l+1, l-k, l+k$ twice each and $2, 2k, 2l$ once each, and a third occurrence needs one of the six coincidences listed. (c) Every configuration lies in a finite part of the hull, and in $S$ no difference occurs three times: for a pair $s < s+d$ in $S$ with $M = \max(|s|, |s+d|)$, either $s = -M$, $d = 2M$, or $|d - M| \leq M/4$ and $s \in \{M-d, -M\}$, and non-zero absolute values in $S$ that differ do so by a factor at least 4. □

**Remark 5.20.** At a candidate, $u^k \in \widetilde{\mathcal{L}}$ brings $u^{-k} = \rho^{-k}\overline{u^k} \in \widetilde{\mathcal{L}}$, so hypotheses on powers of $u$ put Laurent hulls inside $\widetilde{\mathcal{L}}$, and Proposition 5.19(a) says exactly when Roy's theorem refutes them through those hulls. The power hulls of §6 are the case $S = \{0, \pm 1, \pm k\}$. The pairs excluded in (b) are not claimed: each is one substitution in [2] (Corollaire 2(P)(1), Théorème 7(1), Corollaire 5(1)–(2)), which does not state them at a candidate. Part (c) says that the route cannot exclude "$u^{4^j} \in \widetilde{\mathcal{L}}$ for all $j$"; it was not found in the sources read, nor were (a) and parts (a)–(c) of Theorem 5.18, which generalise Theorem 5.14(c), the power hulls of §6 and the generic statements of this section. The linear Cauchy–Davenport inequality is a case of a theorem of Eliahou and Lecouvey, as stated in [23, Theorem 2]; Theorem 5.18(d) uses Baker's theorem [10].

### A decomposition that returns to its start

Splitting the conjecture into cases produces statements that are formally weaker, and it is tempting to read a long chain of such splits as progress. It need not be.

**Proposition 5.21.** *Let* (NT) *be the assertion: for every $u$ with $\mathrm{e}^u$ and $\mathrm{e}^{\bar{u}}$ algebraic, $\operatorname{Re} u \neq 0$, $\operatorname{Im} u \neq 0$, and $\operatorname{Re} u$, $\operatorname{Im} u$, $\operatorname{Re} u / \operatorname{Im} u$ all transcendental, the modulus $|u|$ is transcendental. Then, granting Hermite–Lindemann,* (NT) *is equivalent to Conjecture 1.1.*

*Proof.* Assume (NT) and let $u \neq 0$ have $|u| \in \overline{\mathbb{Q}}$ and $\mathrm{e}^u \in \overline{\mathbb{Q}}$. Then $\mathrm{e}^{\bar{u}} = \overline{\mathrm{e}^u}$ is algebraic. If any of $\operatorname{Re} u$, $\operatorname{Im} u$, $\operatorname{Re} u / \operatorname{Im} u$ is algebraic, or if either of $\operatorname{Re} u$, $\operatorname{Im} u$ vanishes, then $|u|^2 \in \overline{\mathbb{Q}}$ forces $u$ itself algebraic — in each case one coordinate is algebraic and the identity $|u|^2 = (\operatorname{Re} u)^2 + (\operatorname{Im} u)^2$ supplies the other — and Hermite–Lindemann applies. Otherwise (NT) makes $|u|$ transcendental, contradicting the hypothesis. The converse is immediate: a $u$ witnessing the failure of (NT) has algebraic modulus and algebraic $\mathrm{e}^u$. □

(NT) is recorded formally and is open; the equivalence just proved is not itself formalised, and Appendix A marks it so.

(NT) is a natural-looking restriction of the conjecture, reached by imposing genericity on a conjugate pair, and it sits several case splits below the original statement. It is nonetheless the original statement. A decomposition can descend a long way and arrive back where it began, and only closing the cycle detects this; reading the chain forwards never does.

### What is not available

Two standard instruments are inapplicable here for reasons that are structural rather than quantitative, and it is worth recording which.

*Gel'fond–Schneider* produces transcendence of $\alpha^\beta$ from an algebraic $\alpha \neq 0, 1$ and an algebraic irrational $\beta$. Applied to the configuration above it manufactures transcendence at a point unrelated to the one in question; the number whose transcendence is wanted is not of that shape. It is not idle elsewhere: it is what confines the exceptions to $(S)$ to one rational line (Theorem 3.15).

*Baker's theorem* bounds non-vanishing linear forms in logarithms from below. The relevant form here vanishes identically by hypothesis — that is what being a candidate means — so

the bound has nothing to act on. Using it would require knowing in advance that the form is non-zero, which is the conclusion sought.

Finally, the transcendence-degree-one case of four exponentials, which does apply on part of the locus, is a theorem of 1973–74 (Brownawell; Waldschmidt), recorded as Theorem 1 of Roy–Waldschmidt [6]. When this note was first written it was available for citation and for nothing else, and the obstacle to a formal development looked like Philippon's zero estimate together with the apparatus around it. That reading was wrong about the route. Waldschmidt's own proof [13], with the toolbox of his 1971 paper [14], uses none of it: Siegel's lemma, a Gel'fond-type transcendence criterion, the maximum principle, and a zero count for exponential polynomials that needs no separation hypothesis. The last is what lets the argument do without Gel'fond's lower bound for linear forms in two logarithms, which the 1973 paper uses, noting that Baker generalised it, to meet a hypothesis of its Lemme 2 [13, pp. 195, 199]. Along that route the theorem is now formalised, and the case of the branch that it governs is proved with it; see Appendix A. What remains on that branch is not the 1973–74 theorem but two statements about $1/\pi$, both open. The formal statement takes all four exponentials to be algebraic, which is more than the 1973 theorem assumes and exactly what this setting supplies.

### Interpolation on the lattice sees nothing

The classical way to attack $\mathrm{e}^{u}$ and $\mathrm{e}^{\bar{u}}$ together is to run the two-variable exponential system

$$F(z,w)=\exp(uz+\bar{u}\,w)$$

on the lattice $\mathbb{Z}^2$, where its values $F(m,n)=\mathrm{e}^{mu}\mathrm{e}^{n\bar{u}}$ are algebraic, and to feed an auxiliary function built from it to an interpolation or Schneider–Lang argument. Two elementary facts stand in the way.

**Proposition 5.22** (No first-order arithmetic operator)**.** *Let $u$ be a candidate and $X=a\,\partial_z+b\,\partial_w$ with $a,b$ polynomials with algebraic coefficients. If $(XF)(m,n)$ is algebraic for every $(m,n)\in\mathbb{Z}^2$, then $a=b=0$.*

*Proof.* $(XF)(m,n)=\big(a(m,n)u+b(m,n)\bar{u}\big)F(m,n)$, and $F(m,n)$ is a non-zero algebraic number, so $a(m,n)u+b(m,n)\bar{u}$ is algebraic. A candidate lies on neither axis — on an axis, $|u|\in\overline{\mathbb{Q}}$ would make $u$ algebraic, against Hermite–Lindemann — so $u$ and $\bar{u}$ are $\mathbb{Q}$-linearly independent, as in the proof of Theorem 6.1, and Baker's theorem in its inhomogeneous form gives $a(m,n)=b(m,n)=0$. A polynomial in two variables vanishing on $\mathbb{Z}^2$ is zero. □

The same holds for $a,b$ rational functions with algebraic coefficients regular on $\mathbb{Z}^2$, by clearing denominators; the formal statement is the polynomial one. The consequence is that the criterion of Schneider–Lang [17, Theorem 4.1], which asks the derivatives of the functions to stay in a ring generated over a number field, has no input here. No operator of order one has algebraic values on the lattice. The operator that does, $\partial_z\partial_w$, with $\partial_z\partial_w F=u\bar{u}\,F=\rho F$, is exactly where the candidate's relation enters, and it is not a derivation.

**Proposition 5.23** (Kronecker factorisation)**.** *Let $u$ be a candidate and $\alpha=\mathrm{e}^{u}$. For $0\le a,b,m,n\le N$ the matrix $M=\big(\exp(aum+b\bar{u}n)\big)$, with rows indexed by $(a,b)$ and columns by $(m,n)$, is the Kronecker product $A\otimes B$ of $A=(\alpha^{am})$ and $B=(\bar{\alpha}^{\,bn})$. Hence*

$$\det M=(\det A)^{N+1}(\det B)^{N+1}\neq 0.$$

*Proof.* $\exp(aum+b\bar{u}n)=\alpha^{am}\bar{\alpha}^{\,bn}=A_{a,m}B_{b,n}$, which is the Kronecker product, and $\det(A\otimes B)=(\det A)^{N+1}(\det B)^{N+1}$. Each factor is a Vandermonde determinant, in the nodes $\alpha^{a}$ and $\bar{\alpha}^{\,b}$ respectively, and the nodes are pairwise distinct because $|\alpha|=\mathrm{e}^{\operatorname{Re}u}\neq 1$, a candidate having $\operatorname{Re}u\neq 0$. □

So the interpolation determinant of the natural system splits into two independent one-variable determinants, and neither $\rho$ nor the relation $u\bar{u} = \rho$ appears in it. Schwarz' lemma for Cartesian products [17, Proposition 4.7] asks for vanishing on a full simplex of derivatives, which throws the method back on the values; and the values, by Proposition 5.23, see nothing of the candidate's arithmetic.

# 6 Exclusions outside the hull

Section 5 showed that nothing inside $W_u = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$ decides the question: the hull a candidate certifies out of itself is three-dimensional, and the templates that would refute it need four dimensions. The way past that is to leave the hull — to adjoin one further number and ask what it would have to be. This section collects what that yields. Everything here is a necessary condition on a hypothetical counterexample; none of it is an emptiness proof, and the final subsection says where the method stops.

Throughout, $u$ is a candidate, $\rho = |u|^2 \in \overline{\mathbb{Q}}^\times$, and $\bar{u} = \rho/u$, so that $u^{-1} = \bar{u}/\rho$ lies in $\widetilde{\mathcal{L}}$ along with $u$ and 1.

### Algebraic lines, and distance to an algebraic point

**Theorem 6.1.** *Let $\lambda \in \mathcal{L}$ with $\operatorname{Re}\lambda \neq 0$ and $\operatorname{Im}\lambda \neq 0$. There are no $B \in \overline{\mathbb{Q}}^\times$ and $C \in \overline{\mathbb{Q}}$ with*

$$B\lambda + \bar{B}\,\bar{\lambda} + C = 0.$$

*Proof.* Off both axes, $\lambda$ and $\bar{\lambda}$ are $\mathbb{Q}$-linearly independent: taking real and imaginary parts of $p\lambda + q\bar{\lambda} = 0$ with $p, q \in \mathbb{Q}$ gives $(p+q)\operatorname{Re}\lambda = 0$ and $(p-q)\operatorname{Im}\lambda = 0$, whence $p = q = 0$. Both lie in $\mathcal{L}$, since $\mathrm{e}^{\bar{\lambda}} = \overline{\mathrm{e}^{\lambda}}$ is algebraic. Baker's theorem in its inhomogeneous form then makes the non-zero $\overline{\mathbb{Q}}$-combination $B\lambda + \bar{B}\bar{\lambda}$ transcendental, so it cannot equal the algebraic number $-C$. □

The argument is the first step of the proof of [1, Théorème 3].

The condition has a geometric reading. Writing $B = B_1 + iB_2$ and $\lambda = x + iy$, the relation is $2(B_1x - B_2y) + C = 0$: an affine line with real algebraic coefficients, not both zero. So Theorem 6.1 says a logarithm off the axes lies on no such line. For a candidate the same conclusion is available with less input, and it is worth recording which:

**Remark 6.2.** For a candidate $u = x + iy$, no relation $Ax + By = C$ with $A, B, C \in \overline{\mathbb{Q}} \cap \mathbb{R}$ and $(A, B) \neq (0, 0)$ holds, by Hermite–Lindemann alone. Such a line meets the circle $X^2 + Y^2 = \rho$ only in points with algebraic coordinates — solve the linear equation for one coordinate and substitute, the resulting quadratic having non-zero leading coefficient because $A, B$ are real — and $u$ is transcendental. No linear forms in logarithms are needed. This form now has a machine-checked proof of its own; Theorem 6.1 remains the stronger statement, dropping the hypothesis on $|u|$ at the cost of Baker's theorem. It is also the degree-one case of Proposition 6.22 below: a non-zero polynomial of degree at most one cannot be divisible by $X^2 + Y^2 - \rho$.

**Corollary 6.3.** *For every $a \in \overline{\mathbb{Q}}^\times$, $|u - a| \notin \overline{\mathbb{Q}}$. Equivalently, a candidate lies on no circle of algebraic radius about a non-zero algebraic centre. This statement is now machine-checked.*

*Proof.* If $|u - a|$ were algebraic then so would be $2\operatorname{Re}(\bar{a}u) = |u|^2 + |a|^2 - |u - a|^2$, which is a forbidden line by Remark 6.2. The exclusion of $a = 0$ is not cosmetic: $|u|$ is algebraic by hypothesis. □

### The multiplier module

The next statement determines, completely, which numbers can multiply a candidate back into $\widetilde{\mathcal{L}}$. Its input is Roy's strong six exponentials theorem: for $\overline{\mathbb{Q}}$-linearly independent $x_1, x_2$ and $\overline{\mathbb{Q}}$-linearly independent $y_1, y_2, y_3$, at least one of the six products $x_i y_j$ lies outside $\widetilde{\mathcal{L}}$.

**Theorem 6.4.** *Assume the strong six exponentials theorem. For every candidate $u$,*

$$\{z \in \widetilde{\mathcal{L}} :\ uz \in \widetilde{\mathcal{L}}\} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\, u^{-1}. \tag{M}$$

*Proof.* For the inclusion $\supseteq$: $1$ and $u^{-1}$ lie in $\widetilde{\mathcal{L}}$, and for $z = a + bu^{-1}$ with $a, b \in \overline{\mathbb{Q}}$ one has $uz = au + b \in \widetilde{\mathcal{L}}$.

For $\subseteq$, suppose $z \in \widetilde{\mathcal{L}}$, $uz \in \widetilde{\mathcal{L}}$ and $z \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u^{-1}$. Since $u$ is transcendental, $1, u$ are $\overline{\mathbb{Q}}$-linearly independent, and so are $1, u^{-1}$; the assumption on $z$ extends the second pair to an $\overline{\mathbb{Q}}$-linearly independent triple $z, 1, u^{-1}$. The six products of $(1, u)$ with $(z, 1, u^{-1})$ are

$$z, \qquad 1, \qquad u^{-1}, \qquad uz, \qquad u, \qquad u \cdot u^{-1} = 1,$$

all of them in $\widetilde{\mathcal{L}}$. This contradicts the strong six exponentials theorem. □

The inclusion $\subseteq$ is the case $x = (1, u)$, $y = (z, u^{-1})$ of Diaz's Théorème 4(1) [2], and the case $\Lambda_1 = u$ of Waldschmidt's Corollary 2.2 [18].

The force of (M) is that the right-hand side is closed under nothing useful. Two instances:

**Corollary 6.5.** *Assume the strong six exponentials theorem. For every candidate $u$ and all $a \in \overline{\mathbb{Q}}^{\times}$, $b, c \in \overline{\mathbb{Q}}$,*

$$u^2 \notin \widetilde{\mathcal{L}}, \qquad \textit{hence} \qquad au^2 + bu + c \notin \widetilde{\mathcal{L}}.$$

*In particular $\mathrm{e}^{au^2+bu+c}$ is transcendental.*

*Proof.* Take $z = u$ in (M). If $u^2 \in \widetilde{\mathcal{L}}$ then $u = a' + b'u^{-1}$ for some $a', b' \in \overline{\mathbb{Q}}$, so $u^2 - a'u - b' = 0$ and $u$ is algebraic. The second form follows because $\widetilde{\mathcal{L}}$ contains $u$ and $\overline{\mathbb{Q}}$ and is an $\overline{\mathbb{Q}}$-subspace. □

This is Diaz's Corollaire 5(1) [2], since $u/\bar{u} = u^2/\rho$; see also [1, Théorème 3(1)]. His Corollaire 5(2) gives $u^3 \notin \widetilde{\mathcal{L}}$ as well, and his Corollaire 2 (P) 2) gives $\lambda u \notin \widetilde{\mathcal{L}}$ for every transcendental $\lambda \in \widetilde{\mathcal{L}}$ on the real or the imaginary axis [2, p. 381]; at $\lambda = \pi$, $\mathrm{e}^{\beta\pi u}$ is transcendental for every algebraic $\beta \neq 0$, as in [1, Théorème 3(2)]. Diaz's general proofs use Baker's theorem to make $1, \lambda, \bar{\lambda}$ free over $\overline{\mathbb{Q}}$; at a candidate that freeness comes from Hermite–Lindemann, because $\bar{u} = \rho/u$. Both statements are machine-checked, with Roy's theorem as a hypothesis, and so is all of §2 of [2], where they come from: Théorèmes 3 to 7 and Corollaires 1 to 7, the parts that need it with Baker's theorem as a second hypothesis.

No higher power follows this way. For transcendental $u$ and $k \geq 1$, two $\overline{\mathbb{Q}}$-independent $x_i$ and three $\overline{\mathbb{Q}}$-independent $y_j$ with all six products in $\overline{\mathbb{Q}}u^{-k} + \overline{\mathbb{Q}}u^{-1} + \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}u^k$ exist only for $k = 2$ and $k = 3$. After multiplying by $u^k$, the products become polynomials supported on $\{0, k-1, k, k+1, 2k\}$, the rows differ by a fixed shift $\delta \neq 0$ of the orders at 0, and for $k = 1$ and $k \geq 4$ no such shift keeps three exponents of that set inside it. So, through the hull of a single power, Roy's theorem excludes $u^2$ and $u^3$ from $\widetilde{\mathcal{L}}$ and no higher power; pairs of higher powers can still be excluded (Proposition 5.19(b)). Diaz's Théorème 7 gives only $\{u, u^2, u^3\} \not\subset \widetilde{\mathcal{L}}$, and he remarks that inside that theorem one cannot hope to go very far [2, p. 390]; this is one precise instance. The statement was not found in the sources read; it is machine-checked.

**Corollary 6.6.** *Assume the strong six exponentials theorem. For every candidate $u$ and every $a \in \overline{\mathbb{Q}}^{\times}$,*

$$\frac{1}{u - a} \notin \widetilde{\mathcal{L}}.$$

*Proof.* Put $z = (u-a)^{-1}$, a legitimate test element because $uz = 1 + az$, so $uz \in \widetilde{\mathcal{L}}$ as soon as $z$ does. Then (M) gives $z = c + du^{-1}$ with $c, d \in \overline{\mathbb{Q}}$, and clearing denominators,

$$c\,u^2 + (d - ca - 1)\,u - da = 0.$$

The coefficients are algebraic and not all zero: $c = 0$ forces $d = 1$, and then the constant term is $-a \neq 0$. So $u$ is algebraic, a contradiction. □

The contrast is the point. A candidate's reciprocal $1/u$ necessarily lies in $\widetilde{\mathcal{L}}$, while no algebraically shifted reciprocal does. At a candidate, Corollary 6.6 is Diaz's Corollaire 5(4) [2]: $1/\lambda$ and $1/(\lambda + \alpha)$ are not both in $\widetilde{\mathcal{L}}$.

### Commensurable pairs, unconditionally

Corollary 6.5 rests on Roy's theorem, which has no formal proof. Part of it survives on the four exponentials theorem in transcendence degree one alone, through a statement about pairs of logarithms.

**Theorem 6.7.** *Let $u, v \in \mathcal{L} \setminus \{0\}$ with $|u|^2/|v|^2 \in \mathbb{Q}$, and suppose $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}, v, \bar{v}) \leq 1$. Then $v \in \mathbb{Q}u$ or $v \in \mathbb{Q}\bar{u}$.*

*Proof.* With $c = |u|^2/|v|^2$, the matrix $\begin{pmatrix} u & v \\ c\bar{v} & \bar{u} \end{pmatrix}$ has determinant $|u|^2 - c|v|^2 = 0$ and non-zero entries in $\mathcal{L}$. A column relation gives $v \in \mathbb{Q}u$, a row relation $v \in \mathbb{Q}\bar{u}$. □

The hypotheses are satisfiable, by $u = \log 2$ and $v = \log 4$, and both alternatives occur. For a candidate, $\bar{u} = \rho/u$, so the transcendence-degree hypothesis holds for every $v$ algebraic over $\mathbb{Q}(u)$.

**Corollary 6.8.** *Let $u$ be a candidate, $k \geq 2$, and $c \in \overline{\mathbb{Q}}^{\times}$ with $|c|^2\rho^{k-1} \in \mathbb{Q}$. Then $\mathrm{e}^{cu^k}$ is transcendental; in particular $\mathrm{e}^{u^k/|u|^{k-1}}$ is, for every $k \geq 2$.*

*Proof.* If $v = cu^k \in \mathcal{L}$, then $|v|^2 = (|c|^2\rho^{k-1})\rho$, and Theorem 6.7 gives $cu^k = qu$ or $cu^k = q\rho/u$ with $q \in \mathbb{Q}$. Either way a positive power of $u$ is algebraic, so $u$ is algebraic, against Hermite–Lindemann. □

Negative exponents follow by applying the corollary to $\bar{u}$, since $cu^{-k} = (c/\rho^k)\,\bar{u}^k$. The same matrix argument, with $\begin{pmatrix} u & m \\ l & \bar{u} \end{pmatrix}$ for $l = pu + q\bar{u}$ and $m = \rho/l$, shows that $\mathrm{e}^{\rho/(pu+q\bar{u})}$ is transcendental for all non-zero rationals $p, q$. In particular the harmonic mean $\rho/\operatorname{Re} u$ of $u$ and $\bar{u}$ is not in $\mathcal{L}$.

Theorem 6.7 also constrains the candidates that are algebraic over $\mathbb{Q}(\pi)$ and have a rational squared modulus. The points $t + i\pi$ of Theorem 3.5 are of this kind whenever $t^2 + \pi^2$ is rational.

**Corollary 6.9.** *Let $u, v \in \mathcal{L} \setminus \{0\}$ be algebraic over $\mathbb{Q}(\pi)$, with $|u|^2$ and $|v|^2$ rational. Then $v \in \mathbb{Q}u$ or $v \in \mathbb{Q}\bar{u}$.*

*Proof.* The conjugates $\bar{u} = |u|^2/u$ and $\bar{v} = |v|^2/v$ are algebraic over $\mathbb{Q}(\pi)$ as well. Hence the field $\mathbb{Q}(u, \bar{u}, v, \bar{v})$ has transcendence degree at most one over $\mathbb{Q}$, and $|u|^2/|v|^2$ is rational, so Theorem 6.7 applies. □

**Corollary 6.10.** *Let $t_0, t_1$ be real with $\mathrm{e}^{t_0}, \mathrm{e}^{t_1} \in \overline{\mathbb{Q}}$. If $t_0^2 + \pi^2$ and $t_1^2 + \pi^2$ are both rational, then $t_1 = \pm t_0$. In particular $(\log 2)^2 + \pi^2$ and $(\log 3)^2 + \pi^2$ are not both rational.*

*Proof.* The numbers $u = t_0 + i\pi$ and $v = t_1 + i\pi$ lie in $\mathcal{L} \setminus \{0\}$, and they are algebraic over $\mathbb{Q}(\pi)$ because $t_k^2 = (t_k^2 + \pi^2) - \pi^2$. Corollary 6.9 gives $v = qu$ or $v = q\bar{u}$ with $q \in \mathbb{Q}$, and the imaginary parts force $q = \pm 1$. For the example, $\log 3 \neq \pm \log 2$. □

So condition (ii) of Theorem 3.5 can fail at a rational value of $t^2+\pi^2$ for at most one pair $\pm t$. Since $(\log 3)^2-(\log 2)^2=\log\frac{3}{2}\cdot\log 6$, the example also says that $\log\frac{3}{2}\cdot\log 6$ is irrational if $(\log 2)^2+\pi^2$ is rational. Neither corollary was found in the sources read. Each follows in a few lines from the four exponentials theorem in transcendence degree one [13, Corollaire 4], [6, Theorem 1], here through Theorem 6.7; each is also one substitution in Brownawell's Corollary 7 [12, p. 23].

For two candidates the pair theorem becomes a dichotomy, and it decides when two candidates on an axis-parallel line share a modulus. Both results come from the author's unpublished manuscript on the conjecture, where they were derived from [7, Théorème 0.2]. They are direct instances of [13, Corollaire 4].

**Corollary 6.11.** *Let $u, v$ be candidates with $|v|^2/|u|^2\in\mathbb{Q}$. Then $v\in\mathbb{Q}^\times u\cup\mathbb{Q}^\times\bar u$ if and only if $u$ and $v$ are algebraically dependent over $\mathbb{Q}$.*

*Proof.* If $v\in\mathbb{Q}^\times u\cup\mathbb{Q}^\times\bar u$, then $u$ and $v$ are dependent, since $\bar u=\rho/u$. Conversely, if they are dependent, then $\mathbb{Q}(u,\bar u,v,\bar v)$ has transcendence degree at most one, as $\bar u=\rho/u$ and $\bar v=|v|^2/v$, and Theorem 6.7 applies. □

**Corollary 6.12.** *Let $u\neq v$ be candidates with $v-u$ real or purely imaginary. Then the following are equivalent: $|v|^2/|u|^2\in\mathbb{Q}$; $|v|=|u|$; and $v=\bar u$ if $v-u\in i\mathbb{R}$, $v=-\bar u$ if $v-u\in\mathbb{R}$.*

*Proof.* The last two are equivalent by plane geometry, and $|v|=|u|$ gives a rational ratio. Conversely, $v-\bar v=u-\bar u$ or $v+\bar v=u+\bar u$ makes $v$ a root of a quadratic polynomial over $\overline{\mathbb{Q}}(u)$, so $u$ and $v$ are dependent, and Corollary 6.11 gives $v=cu$ or $v=c\bar u$ with $c\in\mathbb{Q}^\times$. A candidate lies on neither axis, so $c=1$ in the first case, which is excluded, and $c=\pm1$ in the second. □

**Theorem 6.13.** *Let $u$ be a candidate and $\mu\in\mathcal{L}\setminus(\mathbb{Q}u\cup\mathbb{Q}\bar u)$. Then* $\operatorname{trdeg}_{\mathbb{Q}}\mathbb{Q}(u,\mu,\mathrm{e}^{u\bar u/\mu})\geq 2$.

*Proof.* Waldschmidt's theorem [13] applies to $x=(u,\mu)$ and $y=(\bar u/\mu,1)$, whose second column gives the algebraic numbers $\mathrm{e}^u$ and $\mathrm{e}^\mu$. Two of the eight numbers are then algebraically independent, and all eight are algebraic over $\mathbb{Q}(u,\mu,\mathrm{e}^{u\bar u/\mu})$. □

When $\mu$ is algebraic over $\mathbb{Q}(u)$, the theorem gives the transcendence of $\mathrm{e}^{\rho/\mu}$, as in the remark after Corollary 6.8. It is [13, Corollaire 4] at $(u,\bar u,\mu)$; see also [15, Exercise 7.4.d] and [7, Corollary 1.2(c)].

## Products on the axes

The matrix of Theorem 6.7 has a sibling that reads the argument of a product instead of a ratio of moduli.

**Proposition 6.14.** *Let $\lambda,\mu\in\mathcal{L}\setminus\{0\}$ with* $\operatorname{trdeg}_{\mathbb{Q}}\mathbb{Q}(\lambda,\bar\lambda,\mu,\bar\mu)\leq 1$. *If $\lambda\mu$ is real, then $\lambda$ and $\mu$ are both real, or both purely imaginary, or $\mu\in\mathbb{Q}\bar\lambda$. If $\lambda\mu$ is purely imaginary, one of $\lambda,\mu$ is real and the other purely imaginary.*

*Proof.* If $\lambda\mu$ is real, the matrix $\begin{pmatrix}\lambda & \bar\lambda\\ \bar\mu & \mu\end{pmatrix}$ has non-zero entries in $\mathcal{L}$ and determinant $\lambda\mu-\overline{\lambda\mu}=0$. A row relation gives $\mu\in\mathbb{Q}\bar\lambda$. A column relation gives $\bar\lambda=q\lambda$ and $\mu=q\bar\mu$, and $|q|=1$ forces $q=\pm1$. If $\lambda\mu$ is purely imaginary, the matrix $\begin{pmatrix}\lambda & \bar\lambda\\ -\bar\mu & \mu\end{pmatrix}$ has determinant $\lambda\mu+\overline{\lambda\mu}=0$; a row relation forces $\mu=0$, and a column relation puts $\lambda$ and $\mu$ on different axes. □

With $\lambda=\bar\ell_0$ and $\mu=\ell_1$, $\lambda\mu=|\ell_0|^2\,\ell_1/\ell_0$, so the product lies on an axis exactly when the quotient does. That turns the proposition into Diaz's conjecture (Qr2) [2, p. 376] in transcendence degree one. Diaz derives (Qr2) from the four exponentials conjecture with the same matrix [2, p. 377].

**Theorem 6.15.** *Let $\ell_0, \ell_1 \in \mathcal{L}$, neither real nor purely imaginary, with* $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(\ell_0, \overline{\ell}_0, \ell_1, \overline{\ell}_1) \le 1$. *If $\ell_1/\ell_0$ is real or purely imaginary, then $\ell_1/\ell_0 \in \mathbb{Q}$.*

For a candidate the hypothesis on the transcendence degree holds for every $\mu$ algebraic over $\mathbb{Q}(u)$, since $\bar{u} = \rho/u$, and $u$ lies on neither axis, by Hermite–Lindemann.

**Corollary 6.16.** *Let $u$ be a candidate and $\mu \in \mathcal{L} \setminus \{0\}$ algebraic over $\mathbb{Q}(u)$. Then $u\mu$ is real if and only if $\mu \in \mathbb{Q}\bar{u}$, and $u\mu$ is never purely imaginary.*

Diaz quotes the four exponentials theorem in transcendence degree one on p. 386 of [2] but does not combine it with (Qr2); his Corollaire 6(2) proves (Qr2) when $\ell_1/\ell_0 \in \widetilde{\mathcal{L}}$ instead. The combination was not found in the sources read. Under Schanuel's conjecture the hypothesis of Theorem 6.15 never holds off the axes, so these statements, like Theorem 6.7, constrain configurations that are not known to be impossible, and Corollary 6.16 is vacuous if Conjecture 1.1 holds.

## Two candidates, and the six exponentials theorem

Theorem 5.3 says the six exponentials theorem cannot refute one candidate: its certificate is three-dimensional. Two candidates whose squared moduli have rational ratio supply the missing dimension.

**Theorem 6.17.** *Let $u, v \in \mathcal{L} \setminus \{0\}$ with $|u|^2/|v|^2 = c \in \mathbb{Q}$ and $v \notin \mathbb{Q}u \cup \mathbb{Q}\bar{u}$. If $w \in \mathcal{L}$ and $uw/v \in \mathcal{L}$, then $w \in \mathbb{Q}v + \mathbb{Q}\bar{u}$.*

*Proof.* Put $m = u/v \notin \mathbb{Q}$. Then $mv = u$ and $m\bar{u} = c\,\bar{v}$ lie in $\mathcal{L}$. If $w \notin \mathbb{Q}v + \mathbb{Q}\bar{u}$, the six products of $x = (1, m)$ and $y = (v, \bar{u}, w)$ all lie in $\mathcal{L}$, with $x$ and $y$ both $\mathbb{Q}$-independent, against the six exponentials theorem. □

Two lemmas turn this into a statement about candidates. If $p, q$ are candidates and $\operatorname{Re}(pq) \in \overline{\mathbb{Q}}$, then $q \in \mathbb{Q}\bar{p}$: indeed $pq$ is then algebraic, since $|pq|^2$ is, and Gel'fond–Schneider applied to $q = (q/\bar{p})\bar{p}$ forces $q/\bar{p} \in \mathbb{Q}$. And if $u \in \mathcal{L}$ lies off both axes and $w$, $uw/\bar{u} \in \mathcal{L}$, then $a|u|^2 + b\,uw + c\,\overline{uw} = 0$ for rationals $a, b, c$ not all zero. That lemma is six exponentials with $x = (1, u/\bar{u})$ and $y = (\bar{u}, w, \overline{(u/\bar{u})w})$, the $\mathbb{Q}$-linear form of Diaz's Corollaire 4(4) [2]; the corollary itself, in $\widetilde{\mathcal{L}}$ and under Roy's theorem, is machine-checked too.

**Theorem 6.18.** *Let $u, v, w$ be candidates with $|u|^2/|v|^2 \in \mathbb{Q}$. If $\mathrm{e}^{uw/v} \in \overline{\mathbb{Q}}$, then $v \in \mathbb{Q}u$, $w \in \mathbb{Q}v$ or $w \in \mathbb{Q}\bar{u}$.*

*Proof.* If $v \in \mathbb{Q}\bar{u}$, the second lemma makes $\operatorname{Re}(uw)$ algebraic, and the first gives $w \in \mathbb{Q}\bar{u}$. Otherwise, if $v \notin \mathbb{Q}u$, Theorem 6.17 gives $w = av + b\bar{u}$, and $|w|^2 = a^2|v|^2 + b^2|u|^2 + 2ab\operatorname{Re}(uv)$. If $ab \neq 0$ this makes $\operatorname{Re}(uv)$ algebraic, and the first lemma puts $v \in \mathbb{Q}\bar{u}$. □

**Corollary 6.19.** *For real algebraic $r > 0$ let $\Phi_r \subset \mathbb{R}/\pi\mathbb{Z}$ be the set of arguments of the candidates of modulus $r$. If $\varphi_1 + \varphi_2 = \varphi_3 + \varphi_4 \neq 0$ with all $\varphi_j \in \Phi_r$, then $\{\varphi_1, \varphi_2\} = \{\varphi_3, \varphi_4\}$. In particular $\Phi_r$ contains no three-term arithmetic progression.*

*Proof.* The relation gives $u_1u_2 = \pm u_3u_4$ for candidates $u_j$ of modulus $r$, so $\mathrm{e}^{u_1u_2/u_3} \in \overline{\mathbb{Q}}$, and Theorem 6.18 applies. The third alternative, $u_2 \in \mathbb{Q}\bar{u}_1$, means $\varphi_1 + \varphi_2 = 0$. □

So the arguments of candidates on a circle form a Sidon set, up to conjugation. Two candidates of commensurable moduli that are not trivially related generate transcendence degree two (Theorem 6.7), where the four exponentials theorem in degree one is silent. For such a pair Diaz's Corollaire 4(2) [2], a consequence of Roy's strong six exponentials theorem, already puts one of $u/v$ and $v/u$ outside $\widetilde{\mathcal{L}}$, since $u$, $v$, $uv$ are then linearly independent over $\overline{\mathbb{Q}}$.

The same two tools reach an exponent that Gel'fond–Schneider cannot. For every candidate $u$,

$$(-1)^{u/\bar{u}} := \mathrm{e}^{i\pi u/\bar{u}} \text{ is transcendental.}$$

If it were algebraic, the second lemma with $w = i\pi$ would give $\pi \operatorname{Im} u \in \mathbb{Q}\rho$, making $u$ algebraic over $\mathbb{Q}(\pi)$; and then the four exponentials theorem in transcendence degree one, applied to $\begin{pmatrix} i\pi u/\bar{u} & i\pi \\ u & \bar{u} \end{pmatrix}$, gives a contradiction, since neither $i\pi \in \mathbb{Q}\bar{u}$ nor a rational relation between $u$ and $\bar{u}$ is possible.

### Saturation at one logarithm

**Theorem 6.20.** *Let $u$ be a candidate and $\ell \in \mathcal{L}$. If $u \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\ell$ then $u \in \mathbb{Q}\ell$.*

*Proof.* Write $u = a + b\ell$ with $a, b \in \overline{\mathbb{Q}}$. If $u$ and $\ell$ are $\mathbb{Q}$-linearly independent, Baker's inhomogeneous theorem makes $u - b\ell$ transcendental, and it equals $a$. So they are $\mathbb{Q}$-dependent: $pu + q\ell = 0$ with $p, q \in \mathbb{Q}$ not both zero. Here $p \neq 0$, since $p = 0$ would give $q\ell = 0$ and hence $\ell = 0$, making $u = a$ algebraic against Hermite–Lindemann. Thus $u = (-q/p)\ell$. □

So inside an algebraic line spanned by one logarithm, the elements of $\mathcal{L}$ are exactly the *rational* multiples of that logarithm: the algebraic span collapses to the rational one. The instance $\ell = i\pi$ closes off a plane one might have expected to have to examine.

**Corollary 6.21.** *No candidate has the form $a + b\pi$ with $a, b \in \overline{\mathbb{Q}}$.*

*Proof.* $i\pi \in \mathcal{L}$, and $a + b\pi = a + (-ib)(i\pi)$ with $-ib$ algebraic. By Theorem 6.20, $u = r\, i\pi$ for some $r \in \mathbb{Q}$, necessarily non-zero; then $|u| = |r|\pi$ is transcendental, contradicting the candidate's algebraic modulus. □

The boundary of this argument should be stated explicitly, because it is easy to over-read. Theorem 6.20 handles one logarithm at a time, with the candidate itself as the second. It says nothing about a combination of two independent logarithms, and in particular it does not touch $\log 2 + i\pi$, a logarithm of $-2$ and the smallest open test case of §3. The distance between the two situations is the whole difficulty.

### Generic on its own circle

One further exclusion belongs with these, and it is the strongest of them. It was the last statement in this note to acquire a machine-checked proof, and the route the formalisation took is not the one given below.

**Proposition 6.22.** *Let $u = x + iy$ be a candidate and $\rho = |u|^2$. For every $P \in \overline{\mathbb{Q}}[X, Y]$,*

$$P(x, y) = 0 \quad \Longleftrightarrow \quad X^2 + Y^2 - \rho \ \ \textit{divides} \ \ P.$$

*Proof.* The implication $\Leftarrow$ is immediate from $x^2 + y^2 = \rho$. For $\Rightarrow$, make the invertible linear change of variables $U = X + iY$, $V = X - iY$ over $\overline{\mathbb{Q}}$, under which $X^2 + Y^2$ becomes $UV$; let $Q \in \overline{\mathbb{Q}}[U, V]$ be the image of $P$, so that $Q(u, \bar{u}) = P(x, y) = 0$ and $\bar{u} = \rho/u$. Modulo $UV - \rho$ the ring $\overline{\mathbb{Q}}[U, V]$ becomes $\overline{\mathbb{Q}}[U, U^{-1}]$, and the residue of $Q$ is a Laurent polynomial vanishing at $u$. Since $u$ is transcendental, a non-zero Laurent polynomial over $\overline{\mathbb{Q}}$ cannot vanish at $u$: clearing denominators would give an algebraic relation for $u$. Hence the residue is zero, $Q \in (UV - \rho)$, and $P \in (X^2 + Y^2 - \rho)$. □

A candidate is therefore algebraically generic on its own circle: it lies on no algebraic plane curve that does not contain that entire circle. It lies on no parabola, and on no ellipse or hyperbola other than through the circle appearing as a component. What the proposition does not do is eliminate the circle itself, which is precisely where the difficulty has been all along — so the statement is sharp in form and empty in consequence for the main question.

The formal proof takes a different route from the one above, and the difference is instructive. Rather than passing to the coordinate ring of the conic, it reads $P$ in $\overline{\mathbb{Q}}[X][Y]$ and divides by the monic quadratic $Y^2 + (X^2 - \rho)$, leaving a remainder $A(X)Y + B(X)$ which must vanish at $(x, y)$. Two facts finish it: $x$ and $y$ are each transcendental — if either were algebraic, $x^2 + y^2 = \rho$ would make the other algebraic and hence $u$ algebraic — and $y \notin \overline{\mathbb{Q}}(x)$, because $\rho - X^2$ is squarefree of degree two and so not a square in $\overline{\mathbb{Q}}(X)$. The change of variables is the shorter argument on paper; the division is the one a proof assistant prefers, because it needs no quotient ring.

# 7 What would have to be proved

Three statements, in increasing strength, each of which settles Conjecture 1.1.

(1) The single relation of Theorem 3.5, for the branch it governs: for real $t \neq 0$ with $\mathrm{e}^t$ algebraic, $t^2 + \pi^2$ is transcendental. Smallest open instance: is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?

(2) The strong five exponentials conjecture of Waldschmidt [16, p. 379], which he notes is weaker than the strong form of the four exponentials conjecture of that paper [16, p. 377], the sharp four exponentials conjecture of [20]. Conjecture 1.2 implies that sharp form with Baker's theorem, so item (2) is indeed weaker than item (3); both steps are machine-checked. With $x_1 = y_1 = 1$ its fifth exponential is one of the four, so the instantiations of Theorems 2.1 and 3.13 apply to it unchanged. It would also give $(\log a)(\log b) \neq \log c$ for non-zero logarithms [16, p. 379], hence the transcendence of $\mathrm{e}^{\pi^2}$ [20]; this too is machine-checked. Version 1.5 put here the rank inequality of §4(a) together with an inhomogeneous extension of Proposition 12.25; the extension is not needed, and for $2 \times 2$ matrices the inequality contains Conjecture 1.2, so it was no weaker than item (3).

(3) The strong four exponentials conjecture, which settles the whole question at once by Theorem 2.1.

Nothing here brings any of the three closer. What the development does establish is where the question is not: not in a better matrix, by Theorem 4.2; not in an algebraic invariant of the candidate's hull, by Theorem 5.1, nor, for a generic candidate, in one algebraic over $\mathbb{Q}(\pi)$, by Theorem 5.2; not in the six exponentials theorem, by Theorem 5.3; and not, on one whole branch, in further case analysis, by Proposition 5.21.

### Where the formal development stops

The case analysis behind this note has been carried to its end. Every branch of Conjecture 1.1 is either closed by a machine-checked proof or reduced, by a machine-checked reduction, to one of exactly two statements, and nothing else in it is open:

(i) the real half of $(S)$: for real algebraic $\gamma \neq 0$, $\mathrm{e}^{-i\gamma/\pi}$ is transcendental;

(ii) the statement (NT) of Proposition 5.21, which is equivalent to the conjecture itself.

Both follow from Conjecture 1.2. Neither is a formalisation problem. Version 1.0 of this note listed a third, the imaginary half of $(S)$: for real algebraic $\beta \neq 0$, $\mathrm{e}^{\beta/\pi}$ is transcendental. It remains open, but the development no longer depends on it, since both branches that reach $(S)$ produce a real $\gamma$. By Theorem 3.15, it and the real half cannot both fail.

It is worth saying why the four exponentials theorem in transcendence degree one, now proved, does not reach them, since each lives in transcendence degree one: under a counterexample to

$(S)$ the numbers $\lambda$ and $i\pi$ are algebraically dependent, $\lambda = \gamma/(i\pi)$, and a candidate's $u$ and $\bar{u}$ are algebraically dependent because $\bar{u} = \rho/u$. The obstruction is the input, not the degree. The configurations that Theorems 2.1 and 3.13 use have products $1,\ \bar{u},\ u,\ \rho$ and $1,\ i\pi,\ \lambda,\ \gamma$, and in each two of the four are non-zero algebraic numbers. A non-zero algebraic number is never in $\mathcal{L}$, by Hermite–Lindemann. The four exponentials theorem, and the four exponentials conjecture above it, ask for their products to be logarithms of algebraic numbers; only the strong form, whose products are allowed to range over $\widetilde{\mathcal{L}}$, accepts these. That distinction, $\mathcal{L}$ against $\widetilde{\mathcal{L}}$, is the whole distance between what is proved and what is needed. Theorem 5.14 and Proposition 5.15 make it precise. On the certified logarithms the four exponentials conjecture is already a theorem, so it can give nothing. What separates $\widetilde{\mathcal{L}}$ from $\mathcal{L}$ here is the constant term, not the algebraic coefficients.

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# A Machine-checked certificates

Every numbered statement above but two has been checked in a formal development, together with Baker's theorem, which four of them use, in Lean 4 against a pinned revision of its mathematical library. The table records where each one lives, so that any claim in this note can be verified rather than taken on trust. The environment is stated here once and nowhere else.

The proofs exist in two places, and the difference matters to a reader who wants to check them. Each identifier below names a node on a collaborative proving platform, where the statement, the submitted proof and its verdict are public. Every proved statement is also mirrored in a self-contained Lean project at

https://github.com/carlok/diaz-modulus-lean

which builds from source in one command, depends on no platform, and assumes nothing beyond Lean's own three axioms. Where the two differ in naming, the project keeps the platform's spelling.

This is version 1.28 of the note; version 1.0 of 21 September 2026 is the first stable release. Version 1.6 follows a reading of the published work of Diaz, Roy, Waldschmidt and Dasgupta–Kakde, and corrects attributions throughout: several results below are classical or special cases of theirs, and each is attributed in the text. Version 1.7 adds rows for the results of that reading that are now machine-checked, and version 1.8 two more: Diaz's Proposition 1 and Waldschmidt's Consequence 1.6. Version 1.9 corrects this appendix against the formal statements. Four rows now say that their proofs take Baker's theorem as a hypothesis, and Section 1 lists it as a third input used without proof; Corollary 3.4 is marked as not formalised; and Proposition 3.2 is restated as its formal counterpart proves it. Version 1.10 withdraws a priority claim in Section 6 that Diaz's Corollaire 4(2) of 2007 contradicts. Version 1.11 adds Corollaries 3.8, 6.9 and 6.10, machine-checked on 29 September 2026, and in Section 4 a consequence of the 1973 theorem, machine-checked with that theorem as a hypothesis. Version 1.12 records that the 1973 theorem itself, Schneider's eighth problem and the $e^{\pi^2}$ statement of Section 4 are machine-checked, on 30 September 2026, so that consequence is now unconditional. Version 1.13 adds, on 1 October 2026, consequences of Roy's strong six exponentials theorem, machine-checked with that theorem as a hypothesis: Diaz's Corollaires 1, 2, 4 and 5 of 2007, and their instances at a candidate and at $i\pi$; and, with Baker's theorem as a second hypothesis, a case of a question in [17, p. 399]. Version 1.12 had left the first sentence of this paragraph at version 1.11. Version 1.14 adds, on the same day, the rest of §2 of Diaz's 2007 paper, the limit of the strong six exponentials route at $u^3$, and the Matrix Coefficient Conjecture on a generic candidate's homogeneous data in every size, with Roy's lemma and the homogeneous-relation statement it uses. Version 1.15 adds, the same day, three consequences of the strong four and strong five exponentials conjectures, machine-checked with those conjectures as hypotheses; corrects item (2) of Section 7, which compared it with the strong four exponentials conjecture of 1988 as though it were Conjecture 1.2; and notes that the homogeneous conjecture on logarithms cannot see a generic candidate either. Version 1.16 adds, on 2 October 2026, Proposition 6.14, Theorem 6.15 and Corollary 6.16: Diaz's (Qr2) in transcendence degree one. Baker's theorem is now machine-checked, so the four rows that assumed it name unconditional forms and read Proved; and the title of Brownawell's 1974 paper is corrected. Version 1.17 adds, on 4 October 2026, Proposition 3.9 and Corollary 3.10 (the dilogarithm $\mathrm{Li}_2(1/2)$ against $e^{i/\pi}$), Theorem 5.4 (no $2 \times 3$ configuration on generic data, whatever is added), and Corollaries 6.11 and 6.12 with Theorem 6.13, three results of the author's manuscript that Waldschmidt's 1973 corollary now gives directly. It credits Brownawell's 1974 corollaries where they apply, and Karatarakis's formalisation of Baker's theorem. Version 1.18 adds, on 5 October 2026, Theorem 5.5: the four-dimensional extensions $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \overline{\mathbb{Q}}z$ that carry a $2 \times 3$ configuration are exactly those containing $u^2$, $\bar{u}^2$ or some $1/(u-a)$. Version 1.19 adds, on 5 October 2026, Remark 2.3, on Kirby's weak form of Schanuel's conjecture, and identifies the $2 \times 2$ case of an extension expected by Dasgupta and Kakde with the sharp four exponentials conjecture with rational constants. Version 1.20 adds, on 6 October 2026, Proposition 5.6: five-dimensional spaces $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u} + \overline{\mathbb{Q}}z + \overline{\mathbb{Q}}\bar{z}$ that carry a $2 \times 3$ configuration invisible to the four-dimensional spaces of Theorem 5.5. Version 1.21 adds, on 8 October 2026, Theorem 5.18 and Proposition 5.19: separation, the normal form of $2 \times 2$ configurations near a point of a circle, and configurations in Laurent hulls; Theorem 4.1 now assumes only $u \notin K$, as its Lean statement does. Version 1.22 adds, on 9 October 2026, Proposition 5.8 (two poles, no conjugation condition), and makes Remark 2.3 formal. Version 1.23 adds, on 10 October 2026, Theorem 5.9 (every involution of the $u$-line) and Remark 5.10 (its converse, not formalised). Version 1.24 adds, on 11 October 2026, Proposition 5.11 (the conjugation-stable spaces of Case II with a factor of degree one), makes the index-one step of Remark 5.10 formal, and adds

the kernel form of Schanuel's conjecture to Remark 2.3. Version 1.25 makes, on 12 October 2026, the classification of Remark 5.10 formal for a ratio rational in $u$. Version 1.26 adds, on 14 October 2026, Proposition 5.12 (configurations of degree two) and Remark 5.13 (no higher degree; visibility for Laurent $f$). Version 1.27 corrects, on 15 October 2026, three attributions found by two referee reports: that a configuration in a four-dimensional space is a progression is Lemma 4 of Bachoc, Serra and Zémor; Fischler's Lemma 6.1 is about logarithms, and Diaz's Théorème 7(2) is its form for $\widetilde{\mathcal{L}}$; and the strong six exponentials route stops at $u^3$ only for single powers through their hulls. Version 1.28 adds, on 15 October 2026, items (iii) and (iv) of Proposition 5.12, so that each family of the degree-two classification in Remark 5.13 has a formal member, and calls the results that the blueprint lists as routine by that name rather than "short". Every row of the table below was checked against the platform by script on 15 October 2026, and the mirrored project held all 418 proved results of the development at that date.

The 1973–74 theorem's row stands for a development rather than a single proof: the criterion of Waldschmidt 1971, the zero count for exponential polynomials, the construction of 1973 with its auxiliary function, extrapolation and norm step, and the assembly above them. They are separate nodes under the `FourExp` namespace on the platform, and separate modules in the project. The 1973 theorem in its column form reuses that development, with twelve further nodes.

Two further statements, not numbered above because the note does not use them directly, are proved alongside Proposition 6.22 and belong to its argument: the real and imaginary parts of a candidate are each transcendental (`DiazModulus.candidate_re_transcendental`, `DiazModulus.candidate_im_transcendental`).

A **Proved** row means the statement itself has a machine-checked proof. Until version 1.15 four rows read **Proved, assuming Baker**: their proofs took Baker's theorem as a hypothesis. Baker's theorem is now machine-checked, and those rows name the unconditional forms. An **Open** row means the statement is recorded formally but not proved; every such row here is an open conjecture. A **Not formalised** row means what it says: the statement is proved in this note and has no formal counterpart at all. Corollary 3.4 and Proposition 5.21 are flagged that way. The row of Proposition 5.21 needs a word: the identifier given is the statement (NT), which is recorded formally and is Open; the *equivalence* proved in Proposition 5.21 is what has no formal counterpart.

| **Result** | **Status** | **Identifier** |
|---|---|---|
| Conj. 1.1 | Open (conjecture) | `DiazModulus.diaz_modulus_conjecture` |
| Thm 2.1 | Proved | `DiazModulus.diaz_of_sfe` |
| §2, $\mathrm{e}^{u\lambda}$ for $u \in \widetilde{\mathcal{L}}$ | Proved | `DiazModulus.algebraic_modulus_log_multiplier_rational` |
| §2, $\mathrm{e}^{u\lambda}$, every $u$, from Conj. 1.2 | Proved | `DiazModulus.algebraic_modulus_log_multiplier_rational_of_sfe` |
| Rem. 2.2 | Proved | `DiazModulus.diaz_of_schanuel` |
| Thm 3.1 | Proved | `Diaz.real_quantisation` |
| Prop. 3.2 | Proved | `Diaz.quantisation_orbit_iff_re_ne_zero` |
| Thm 3.3 | Proved | `Diaz.fibre_at_most_two` |
| Cor. 3.4 | Not formalised | none; it follows from the identity `Diaz.period_plane_norm` |
| Thm 3.5 | Proved | `DiazModulus.leaf_iff_one` |
| Thm 3.6 | Proved | `DiazModulus.diaz_number_forces_transcendence` |
| Cor. 3.7 | Proved | `DiazModulus.log_two_diaz_or_transcendental` |
| Cor. 3.8 | Proved | `DiazModulus.recip_pi_log_of_torsion_rational_modulus` |
| Prop. 3.9 | Proved | `DiazModulus.recip_pi_log_of_rational_quadratic_relation` |

| Result | Status | Identifier |
|---|---|---|
| Cor. 3.10 | Proved | `DiazModulus.dilog_half_irrational_or_exp_i_div_pi_transcendental` |
| §3, Waldschmidt's remark | Proved | `DiazModulus.log_pair_square_ratio_transcendental`, `DiazModulus.log_two_pi_dependent_forces_transcendence` |
| §3, Diaz's Proposition 1 | Proved | `DiazModulus.log_pair_algebraicIndependent_of_mul_eq_rat_pi_sq` |
| Thm 4.1 | Proved | `Diaz.no_vanishing_coeff` |
| Thm 4.2 | Proved | `Diaz.det_pencil_eq_conic` |
| Prop. 4.3 | Proved | `Diaz.roy_conic_implies_empty` |
| §4, the $2 \times 2$ conjecture | Proved | `DiazModulus.diaz_of_two_by_two_determinant_conjecture` |
| §4, the companion $e^{\lvert\lambda\rvert}$ | Proved | `DiazModulus.exp_abs_transcendental_of_conj_algebraic`, `DiazModulus.exp_abs_log_two_add_i_pi_transcendental` |
| Thm 5.1 | Proved | `Diaz.candidate_indistinguishable_by_coeff` |
| §5, non-candidates on every circle | Proved | `DiazModulus.exists_noncandidate_transcendental_on_circle` |
| Thm 5.2 | Proved | `DiazModulus.generic_indistinguishable_over_pi`, `DiazModulus.circle_points_indistinguishable` |
| Thm 5.3 | Proved | `DiazModulus.sixExponentials_cannot_refute_candidate` |
| Thm 5.4 | Proved | `DiazModulus.generic_circle_point_no_two_by_three_configuration` |
| Thm 5.5 | Proved | `DiazModulus.circle_point_extension_two_by_three_configuration_iff`, `DiazModulus.two_by_three_configuration_forces_progression`, `DiazModulus.cubic_product_eq_square_normal_form`, `DiazModulus.circle_hull_progression_contains_square_or_reciprocal`, `DiazModulus.circle_point_extension_carries_two_by_three_configuration` |
| Prop 5.6 | Proved | `DiazModulus.circle_point_conjugate_pair_configuration_invisible_to_four_dimensional_extensions`, `DiazModulus.circle_point_conjugate_pair_extension_carries_two_by_three_configuration`, `DiazModulus.circle_point_conjugate_pair_extension_excludes_squares_and_reciprocals`, `DiazModulus.conj_stable_circle_point_extension_no_two_by_three_configuration`, `DiazModulus.candidate_div_sq_sub_not_mem_logAlgTilde` |

| Result | Status | Identifier |
|---|---|---|
| Thm 5.18 | Proved | DiazModulus.rank_one_config_separation_two, DiazModulus.rank_one_config_separation, DiazModulus.polynomial_submodule_trailing_degrees_card, DiazModulus.polynomial_submodule_mul_finrank_ge, DiazModulus.rank_one_config_card_le_finrank_add_one, DiazModulus.circle_point_config_card_le_four, DiazModulus.circle_point_two_by_two_normal_form, DiazModulus.generic_circle_point_config_is_two_by_two, DiazModulus.circle_point_config_constant_matrix_det_ne_zero, DiazModulus.algebraic_mem_span_logAlg_eq_zero, DiazModulus.candidate_two_by_two_config_entry_not_mem_span_logAlg |
| Prop 5.19 | Proved | DiazModulus.laurent_hull_config_iff, DiazModulus.two_sumset_iff_difference_count, DiazModulus.power_pair_difference_count_iff, DiazModulus.power_pair_hull_two_by_three_iff, DiazModulus.four_pow_hull_no_two_by_three, DiazModulus.candidate_power_pair_not_both_mem_logAlgTilde, DiazModulus.two_by_two_config_critical_progression |
| Prop 5.8 | Proved | DiazModulus.circle_point_two_pole_extension_carries_two_by_three_configuration, DiazModulus.circle_point_two_pole_extension_excludes_squares_and_reciprocals, DiazModulus.circle_point_two_pole_configuration_invisible_to_four_dimensional_extensions, DiazModulus.candidate_two_pole_not_both_mem_logAlgTilde, DiazModulus.candidate_sum_div_sq_sub_not_mem_logAlgTilde |
| Rem 2.3 | Proved | DiazModulus.candidate_im_mem_pi_rat_of_weak_schanuel, DiazModulus.diaz_iff_single_relation_of_weak_schanuel, DiazModulus.diaz_of_weak_schanuel_and_kernel_schanuel |
| Thm 5.9 | Proved | DiazModulus.involution_pencil_space_carries_two_by_three_configuration, DiazModulus.involution_pencil_space_excludes_squares_and_reciprocals, DiazModulus.involution_pencil_configuration_invisible_to_four_dimensional_extensions |
| Rem 5.10 | Proved | DiazModulus.laurent_case_two_configuration_invisible, DiazModulus.mobius_progression_space_contains_square_or_reciprocal, DiazModulus.progression_space_ratio_degree_le_two, DiazModulus.quadratic_ratio_progression_space_eq_involution_pencil_space, DiazModulus.progression_space_contains_square_or_reciprocal_or_eq_involution_pencil_space |

| Result | Status | Identifier |
|---|---|---|
| Prop 5.11 | Proved | DiazModulus.conj_stable_real_pencil_space_carries_two_by_three_configuration, DiazModulus.conj_equivariant_degree_one_space_eq_real_pencil_space, DiazModulus.real_pencil_quartic_space_excludes_squares_and_reciprocals, DiazModulus.real_pencil_quartic_configuration_invisible_to_four_dimensional_extensions, DiazModulus.candidate_real_pencil_not_both_mem_logAlgTilde |
| Prop 5.12 | Proved | DiazModulus.laurent_cubic_space_excludes_squares_and_reciprocals, DiazModulus.laurent_cubic_configuration_invisible_to_four_dimensional_extensions, DiazModulus.laurent_cubic_space_no_mobius_ratio_configuration, DiazModulus.two_orbit_space_excludes_squares_and_reciprocals, DiazModulus.two_orbit_configuration_invisible_to_four_dimensional_extensions, DiazModulus.conj_stable_two_orbit_space, DiazModulus.conj_swapped_two_orbit_configuration_invisible_to_four_dimensional_extensions, DiazModulus.cyclic_pair_space_excludes_squares_and_reciprocals, DiazModulus.cyclic_pair_configuration_invisible_to_four_dimensional_extensions, DiazModulus.cyclic_pair_space_no_mobius_ratio_configuration |
| Rem 5.13 | Proved | DiazModulus.quadratic_over_ratio_of_mem_progression_pair_span, DiazModulus.progression_pair_span_ratio_degree_le_two, DiazModulus.laurent_real_pencil_space_excludes_squares_and_reciprocals, DiazModulus.laurent_real_pencil_configuration_invisible_to_four_dimensional_extensions, DiazModulus.laurent_real_pencil_space_contains_square_of_root |
| Thm 5.14 | Proved | DiazModulus.generic_conj_pair_no_quadratic_relation, DiazModulus.generic_conj_pair_four_exp_barrier, DiazModulus.generic_qbar_homogeneous_four_exp_barrier |
| Prop. 5.15 | Proved | DiazModulus.candidate_nongeneric_four_exp_barrier |
| Thm 5.16(a) | Proved | DiazModulus.generic_quadratic_relation_is_norm |
| Thm 5.16(b) | Proved | DiazModulus.generic_period_never_enters |
| Thm 5.16(c) | Proved | DiazModulus.generic_no_strong_six_exp_configuration |
| Prop. 5.17(a) | Proved | DiazModulus.conj_pair_quadratic_relation_iff |
| Prop. 5.17(b) | Proved | DiazModulus.anisotropic_relation_four_exp_barrier, DiazModulus.det_linear_forms_isotropic |
| Prop. 5.17(c) | Proved | DiazModulus.anisotropic_relation_on_circle |

| **Result** | **Status** | **Identifier** |
|---|---|---|
| §5, the barrier for $(S)$ | Proved | `DiazModulus.recip_pi_log_four_exp_barrier` |
| §5, $2 \times 2$ matrices of linear forms | Proved | `DiazModulus.det_zero_linear_forms_rank_one`, `DiazModulus.det_zero_linear_forms_rank_one_field` |
| Prop. 5.21 | Not formalised | `DiazModulus.norm_transcendental_of_generic_conj_pair` (Open) |
| §5, 1973–74 thm | Proved | `DiazModulus.four_exponentials_trdeg_one` |
| Prop. 3.11 | Proved | `DiazModulus.recip_pi_log_of_period_aligned` |
| Conj. 3.12 | Open (conjecture) | `DiazModulus.recip_pi_not_log` |
| Thm 3.13 | Proved | `DiazModulus.recip_pi_not_log_of_sfe` |
| §3, an exception to $(S)$ | Proved | `DiazModulus.pi_powers_not_both_mem_logAlgTilde` |
| After Thm 3.13, Consequence 1.6 | Proved | `DiazModulus.div_not_mem_logAlgTilde_of_sfe` (it states Consequence 1.7, which contains 1.6) |
| §7, item (2) | Proved | `DiazModulus.diaz_of_strong_five_exponentials`, `DiazModulus.recip_pi_not_log_of_strong_five_exponentials` |
| §7, item (2) from (3) | Proved | `DiazModulus.sharp_four_and_strong_five_exponentials_of_strong_four` |
| §7, item (2), $e^{\pi^2}$ | Proved | `DiazModulus.exp_log_mul_log_transcendental_of_strong_five` |
| Prop. 3.14 | Proved | `DiazModulus.recip_pi_exp_value_not_root_of_unity` |
| Thm 3.15(a) | Proved | `DiazModulus.recip_pi_log_rational_line` |
| Thm 3.15(b) | Proved | `DiazModulus.recip_pi_log_on_axis` |
| Thm 3.15(c) | Proved | `DiazModulus.recip_pi_not_log_real_or_imag` |
| §3, $\pi \log 2$ or $\pi \log 3$ | Proved | `DiazModulus.pi_log_two_or_pi_log_three_transcendental` |
| §3, partners of $e^{i\gamma/\pi}$ | Proved | `DiazModulus.recip_pi_or_pi_cube` |
| §3, $e^{i/\pi}$ or $e^{i\pi^3}$ | Proved | `DiazModulus.exp_i_div_pi_or_exp_i_pi_cube_transcendental` |
| §3, real half of $(S)$ | Open (conjecture) | `DiazModulus.recip_pi_not_log_real_gamma` |
| §3, imaginary half of $(S)$ | Open (conjecture) | `DiazModulus.recip_pi_not_log_imag_gamma` |
| Prop. 5.22 | Proved | `DiazModulus.no_first_order_arithmetic_operator_unconditional` |
| Prop. 5.23 | Proved | `DiazModulus.kronecker_factorisation` |
| Prop. 4.4 | Proved | `DiazModulus.geometric_triple_not_logs` |
| Cor. 4.5 | Proved | `DiazModulus.exp_pi_sq_or_exp_i_pi_cube_transcendental` |
| §4, $\pi^2$ and $\pi^3$ | Proved | `DiazModulus.pi_powers_not_both_mem_logAlgTilde` |
| §4, the 1973 theorem | Proved | `DiazModulus.two_algebraically_independent_of_exp_column` |
| §4, $e^{\pi^2}$ algebraic | Proved | `DiazModulus.algebraicIndependent_e_pi_of_exp_pi_sq_algebraic` |
| §4, Schneider's problem | Proved | `Transcendence.exp_e_or_exp_e_sq_transcendental` |
| §4, $e^{ir/\pi}$ | Proved | `DiazModulus.pi_e_exp_pi_sq_indep_of_exp_i_rat_div_pi_algebraic` |
| §5, $2^\pi$, $3^\pi$ or $5^\pi$ | Proved | `DiazModulus.two_three_five_pow_pi_transcendental` |
| §5, $e^{\lambda^2/\mu}$ or $e^{\mu^2/\lambda}$ | Proved | `DiazModulus.log_square_duality` |
| §5, $2^{\log_3 2}$ or $3^{\log_2 3}$ | Proved | `DiazModulus.two_pow_log_three_or_three_pow_log_two` |

| Result | Status | Identifier |
|---|---|---|
| Thm 6.1 | Proved | DiazModulus.no_algebraic_generalized_line_unconditional |
| Rem. 6.2 | Proved | DiazModulus.candidate_no_real_algebraic_line |
| Cor. 6.3 | Proved | DiazModulus.candidate_distance_transcendental |
| Thm 6.4 | Proved | DiazModulus.candidate_multiplier_module |
| Cor. 6.5 | Proved | DiazModulus.candidate_multiplier_module |
| Cor. 6.6 | Proved | DiazModulus.candidate_multiplier_module |
| §6, $u^3$ and $\lambda u$ | Proved | DiazModulus.candidate_cube_and_axis_multiple_not_mem_logAlgTilde |
| §6, no higher power | Proved | DiazModulus.power_hull_strong_six_exp_configuration_iff |
| §6, [2], Th. 3, Cor. 1, 2, 4, 5 | Proved | DiazModulus.strong_six_exponentials_iff_quotient_form, DiazModulus.diaz_2007_cor1_PQ, DiazModulus.diaz_2007_cor2_P_consequences, DiazModulus.diaz_2007_cor4, DiazModulus.diaz_2007_cor5 |
| §6, [2], Th. 4–7, Cor. 3, 6, 7 | Proved | DiazModulus.diaz_2007_th4, DiazModulus.diaz_2007_th5, DiazModulus.diaz_2007_th6, DiazModulus.diaz_2007_th7, DiazModulus.diaz_2007_cor3_Q, DiazModulus.diaz_2007_cor6, DiazModulus.diaz_2007_cor7 |
| Thm 6.7 | Proved | DiazModulus.log_pair_rigid_of_trdeg_one |
| Cor. 6.8 | Proved | DiazModulus.candidate_monomial_not_log |
| §6, harmonic mean | Proved | DiazModulus.candidate_harmonic_not_log |
| Cor. 6.9 | Proved | DiazModulus.nongeneric_rational_modulus_orbit |
| Cor. 6.10 | Proved | DiazModulus.torsion_rational_modulus_unique, DiazModulus.log_two_log_three_not_both_rational |
| Cor. 6.11 | Proved | DiazModulus.candidate_pair_dichotomy |
| Cor. 6.12 | Proved | DiazModulus.candidate_axis_ratio |
| Thm 6.13 | Proved | DiazModulus.candidate_mixed_rigidity |
| Prop. 6.14 | Proved | DiazModulus.log_mul_real_trichotomy_of_trdeg_one, DiazModulus.log_mul_imaginary_mixed_of_trdeg_one |
| Thm 6.15 | Proved | DiazModulus.diaz_2007_qr2_of_trdeg_one |
| Cor. 6.16 | Proved | DiazModulus.candidate_log_mul_real_iff_rat_conj |
| Thm 6.17 | Proved | DiazModulus.log_ratio_multipliers |
| §6, the two lemmas | Proved | DiazModulus.candidate_conj_product_rational, DiazModulus.conj_ratio_multiplier_relation |
| Thm 6.18 | Proved | DiazModulus.candidate_quotient_rigid |
| Cor. 6.19 | Proved | DiazModulus.candidate_product_relation_trivial |
| §6, $(-1)^{u/\bar{u}}$ | Proved | DiazModulus.candidate_neg_one_pow_ratio |
| Thm 6.20 | Proved | DiazModulus.candidate_one_log_saturation_unconditional |
| Cor. 6.21 | Proved | DiazModulus.candidate_one_log_saturation_unconditional |
| Prop. 6.22 | Proved | DiazModulus.candidate_vanishing_ideal |

| Result | Status | Identifier |
|---|---|---|
| Hermite–Lindemann | Proved | `DiazModulus.hermite_lindemann_holds` |
| Baker's theorem | Proved | `Schanuel.baker_linear_forms_in_logarithms`, `DiazModulus.baker_two_logs` |

Two entries need a word. Hermite–Lindemann is used freely in the text as classical; it is listed because the library used does not contain it, so it had to be established rather than cited. The same holds for Gel'fond–Schneider, which is proved on the platform in another mission, from the formalisation of Karatarakis and Wiedijk, and is mirrored in the project. Baker's theorem is listed for the same reason [10]; it is proved in this mission, along Chapter 4 of [17], and the board's own statement of it is the node closed there. M. Karatarakis has formalised the same route, for every $d_0$, on a public branch of his Mathlib fork [5]. And the statement (NT) of Proposition 5.21 was contributed by another participant in the same development, independently of the work reported here; the equivalence with Conjecture 1.1 is what is established above.

Some results in the development are not represented here at all. Roughly a third of it consists of closure lemmas, coercions and bookkeeping with no content at this level of exposition, and a further group restates in one set of coordinates what another states in a different one. Neither belongs in a paper, and neither is listed.